

The diachrony of iterativity: Evidence from Germanic *r*-verbs

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Abstract

This article examines the diachrony of Germanic iterative “(V)*r*-verbs” (e.g., German *flattern*, English *flutter*) and related deverbal suffixes (as in, e.g., German *folgern* ‘to infer’). While these verbs form a distinct class in North-West Germanic—evidenced by cognates like Old High German *flogarōn* and Old Norse *flogra*—their origin and the development of their iterative semantics remain poorly understood. To clarify the origin of this class, we present a systematic corpus study of four early Germanic languages: Gothic, Old High German, Old English, and Old Norse. We analyze the derivational bases, Aktionsart, and argument structures of early Germanic (V)*r*-verbs, identifying distinct denominal, deadjectival, and deverbative subclasses. Building on the insight that verbalizers diachronically often emerge from nominal or adjectival material through reanalysis in cross-categorical derivation (Grestenberger 2023), we argue that iterative meaning originated in deadjectival verbs derived from eventive roots. In these cases, the (V)*r*-suffix was reanalyzed first as a root modifier and subsequently as a verbalizer expressing event-internal pluractionality. We contrast this class with deadjectival verbs derived from Property Concept roots, which typically yield degree achievement/fientive-factitive alternation verbs expressing a Change of State. Our findings underscore the necessity of integrating morphological theory with empirical, corpus-based analysis to decode morphosemantic change in the diachrony of verbal morphology.

Keywords

iterativity, derivational morphology, morphosemantics, reanalysis, cross-categorical derivation

1 Introduction

1.1 Background: Aktionsart & verb formation in German(ic)

Modern Germanic languages mostly express argument and event structure alternations via prefixation or preverbs, as in present-day German (PDG) *malen* ‘paint’ (atelic) → *be-malen* (telic), *schlagen* ‘hit’ (activity, atelic) → *er-schlagen* (telic, accomplishment), etc. (Henzen 1957: 13-15; Elsen 2011: 212-218; Fleischer & Barz 2012: 378). However, PDG also has a few Aktionsart-modifying suffixes whose diachrony has not received much attention so far, such as diminutive/iterative *-(e)l-* (e.g., *köch-el-n* ‘to simmer’ ← *koch-en* ‘to cook, boil’, *dräng-el-n* ‘to shove’ ← *dräng-en* ‘to push’),¹ factitive/ornative *-ig-* (e.g., *rein-ig-en* ‘to clean’ ← *rein* ‘clean’), and iterative *-(e)r-* (e.g., *flacker-n* ‘to flicker’, *flatter-n* ‘to flutter’). The goal of this article is to investigate how the latter class arose and developed in the older Germanic languages and how the suffix acquired this specifically *iterative* flavor. To contextualize this function of the suffix, consider its behavior in PDG, where *-er-* appears to act as a verbal stem-forming suffix in three major subclasses of verbs:

¹ See Weidhaas & Schmid (2015); Audring, Booij & Jackendoff (2017); Grestenberger & Kallulli (2019); Wahl (2024) on verbal *-(e)l-*, whose modern uses in many ways parallel that of *-(e)r-*, although it is much more widespread and productive than the latter. The diachrony of verbal *-(e)l-* is understudied; the common wisdom is that it is historically related to/derived from the *nominal* diminutive suffix *-(e)l-* (Wilmanns 1899: 96; Wissmann 1932: 27-28; Krahe & Meid 1969b: 263-264; Paul 1959: 20; Grestenberger & Kallulli 2019), which is thus very different from what we propose here for *-(e)r-*. However, like *-(e)r-*, *-(e)l-* also forms *deverbal* iteratives early on whose diachrony is not entirely clear and may need to be revisited. This is outside the scope of this paper, though it is clear that there is more than one diachronic path towards verbal iterativity/pluractionality (cf. also Audring, Leufkens & van Lier 2021).

(1) PDG *er*-verbs

- a. Denominal verbs from nouns with a stem or (old/synchronic) pl. in *-er*, e.g.,
 - i. *ver-gött-er-n* ‘to treat as a god’ (*Gott*, pl. *Gött-er*)
 - ii. *blätt-er-n* ‘to leaf through’ (*Blatt* ‘leaf, page’, pl. *Blätt-er*)
 - iii. *hämmer-n* ‘to hammer’ (*Hammer* ‘hammer’)
- b. Deadjectival verbs derived from comparatives and positives in *-er*, e.g.,
 - i. *mild-er-n* ‘to make mild’ (*mild*, comp. *mild-er*)
 - ii. *ver-größ-er-n* ‘to increase, enlarge’ (*groß*, comp. *größ-er* ‘large, big’)
- c. Deverbal/root-based verbs (including monomorphemic stems), e.g.,
 - i. *flatter-n* ‘to flutter’, *flacker-n* ‘flicker’
 - ii. *folg-er-n* ‘to infer, deduce’ (*folg-en* ‘to follow’)
 - iii. *steig-er-n* ‘to increase’ (*steig-en* ‘to rise, increase’)

The suffix attaches after the root or stem of the derivational base and before the infinitival ending *-(e)n* in these examples (as usual, we use the infinitive as the citation form). While the *-er-* of classes (1a) and (1b) is arguably contained in the derivational base, this is not the case for class (1c), in which *-er-* acts as an iterative verbal stem-forming suffix according to the standard handbooks (e.g., Wilmanns 1899: 93-96; Henzen 1957: 224; Krahe & Meid 1969b: 264; Birkhan 1985: 184). Cognates for this class can be found at least as far back as the common North-West Germanic (NW Germanic) stage (cf., e.g., OHG *flogarōn* ‘to fly around, flutter’; ON *flogra* ‘to flutter; tumble’).

The origin and diachrony of the iterative semantics of class (1c) are not entirely clear. The handbooks cited above generally propose that positive adjectives formed with the (Proto-)Germanic suffix **-ar-* (and in NW-Germanic probably also some comparative forms) were resegmented in the context of deadjectival verb derivation, giving rise to new complex verbal suffixes, e.g.:

(2) OHG *wach-ar* ‘awake’ : $[[wach-ar]_{A-ōn}]_V$ ‘to be(come) awake, vigilant’ $\rightarrow [[wach]_{V/V-}arōn]_V$

However, deadjectival verbs derived from qualitative or Property Concept (PC) adjectives generally form degree achievement verbs of the factitive-fientive alternation (e.g., PDG *sauber* ‘clean’ $\rightarrow$ fact. *säuber-n* ‘to clean’; *mild*, comp. *mild-er*: *mild-er-n* fact. ‘to make mild(er), mitigate’; Engl. *flat*: *to flatten*, etc.) rather than iteratives, from which they differ in a number of ways: while iteratives are atelic activity verbs, degree achievements can be both telic and atelic (Kennedy & Levin 2008; Bobaljik 2012; Beavers & Koontz-Garboden 2020; Fábregas 2023). Moreover, the derivational base differs: degree achievements are derived from PC roots or stems (e.g., Hale & Keyser 1998, 2002; Harley 2005; Koontz-Garboden 2005; Kennedy & Levin 2008; Bobaljik 2012; Beavers & Koontz-Garboden 2020), in other words, bases that express a *state*, while the iterative class in (1c) seems to take verbal roots or stems, hence *events*, as its derivational base. As far as we know, this morphosemantic problem has not been addressed in the literature so far. In this paper, we argue that this is precisely where morphological theory can help make sense of complex historical data. However, in order to achieve this, a robust empirical basis for generalizations is needed, and there is no corpus study so far to confirm the connection between Germanic *r*-iteratives and *r*-adjectives proposed in the handbooks. While we ultimately expect the handbooks to be correct in this regard, a corpus study provides the basis for both a quantitative and a qualitative evaluation of the available material and therefore the potential for more fine-grained results, especially regarding the question of what role (if any) comparatives in *-r* played in the diachronic development of verbal *-r*.

1.2 Goals and outline of this article

In this article, we present the results of a corpus study of four older Germanic languages to confirm the proposed connection between adjectival *(*)-(a)r-* and verbal iterative *(*)-(a)r-*. Based on this study,

we argue that there is a systematic connection between the semantics of the derivational base and factitive-fientive vs. iterative verb formation in our corpus. Our core claim is that iteratives are not derived from canonically gradable Property Concept adjectives but derivationally associated with so-called “verbal adjectives”: adjectives built from eventive *roots*. Such adjectives differ from, e.g., participles, which are based on the verbal *stem* and include stem-forming morphology, as well as displaying typically “verbal” properties such as the ability to occur with direct case objects if formed to transitive verbs. Verbal adjectives, on the other hand, are *root*-derived. Here, we argue that the type of root on which these adjectives are based matters not only for the semantics of the respective adjectives, but also for that of the verbs that are then derived from such adjectives. This means that the “deadjectival” origin of verbal iterative *-(V)r-* in NW-Germanic is essentially the correct analysis, but that only a specific semantic class of adjectives became the input to this diachronic reanalysis. We further argue that this type of change constitutes evidence for the view that strict locality conditions such as those proposed by, e.g., Bobaljik (2000, 2012) hold between adjacent morphemes – morphosyntactic terminal nodes – diachronically as well as synchronically, in that both the morphological and the semantic content of such nodes are “inherited” by the adjacent higher node in the course of the reanalysis (Grestenberger 2023).

This article is structured as follows. In Section 2, we provide an overview of the relevant verb classes that contain verbal stem-forming *-r-* in the older Germanic languages and provide some background on their prehistory and their inflectional properties. In Section 3, we discuss the methodology of the corpus study. Section 4 presents the results of the study, namely for all *r*-verbs in the selected languages (Section 4.1), the verbs in which *-r-* is part of the base (Section 4.2), the verbs in which *-r-* is *not* part of the base (Section 4.3), as well as a summary of the Aktionsart properties of these classes (Section 4.4). Section 5 presents our analysis using Distributed Morphology (DM), beginning with an introduction to cross-categorical derivation in diachrony (5.1) and an overview of Germanic adjective classes formed with the suffix **-r(a)-* (5.2). We then present arguments against a potential analysis of the iterative class as diachronically derived from deadjectival degree achievement verbs based on PC adjectives, from which the iterative class differs semantically, syntactically, and (at least in OHG) also morphologically (Section 5.1). In Section 5.3, we argue that the iterative class instead has its diachronic source in deadjectival verbs from verbal adjectives, i.e., adjectives derived from roots denoting an event and propose a tentative explanation for the origin of the iterative (event-internal pluractional) semantics of this class. Section 6 concludes.

Before proceeding, a note on the terminology used in this article is in order. We use “root” in the DM sense, i.e., categorially unspecified open class lexical item, while a “stem” minimally consists of a root + a category-determining stem-forming suffix or “categorizer”. “Base” refers to the input for word formation and encompasses both roots and stems, depending on the type of word formation process. We moreover use the terms *fientive* and *factitive* for (respectively) intransitive and transitive change of state verbs based on adjectival *states* (PC lexemes), i.e., degree achievement verbs, as opposed to *inchoative* and *causative*, which we reserve for the intransitive and transitive alternants of verbs derived from *eventive* bases, i.e., causative alternation verbs. We thereby follow the arguments made by, e.g., Koontz-Garboden (2005, 2014), Beavers & Koontz-Garboden (2020), Beavers et al. (2021), who have shown that these two classes systematically differ with respect to their morphology and semantics across a wide sample of genetically unrelated languages. While verbs based on adjectives (adjectival states) form both basic and derived statives (= primary and secondary/deverbal adjectives), verbs based on eventive or “result roots” (in Beavers et al.’s terminology) can only form derived statives which always entail a change of state. This difference is illustrated in Table 1 for English. Note that result statives always entail a change of state due to a prior event, while primary statives do not entail a prior event.

a.	Adjectival root	Basic/primary stative	Factitive/fientive	Result stative
	<i>flat</i>	<i>flat</i>	<i>flatten</i>	<i>flatten-ed</i>
	<i>red</i>	<i>red</i>	<i>redde</i>	<i>redde-ed</i>
b.	Eventive/result root	Basic/primary stative	Causative/inchoative	Result stative
	<i>crack</i>	—	<i>crack</i>	<i>crack-ed</i>
	<i>cook</i>	—	<i>cook</i>	<i>cook-ed</i>

Table 1: Factitive/fientive and causative/inchoative verbs and their bases

Since the difference between state- and event-derived adjectives (basic vs. result statives, in Beavers et al.'s terminology) will be crucial to our argument in Section 5, we therefore also use distinct terms for referring to verbs derived from bases designating states vs. events.

We moreover use *r-verb* and *r-adjective* as cover terms for the different allomorphs and synchronic and diachronic variants of these stem-forming segments (e.g., *-(a)r-* vs. *-r(a)-*) in the various older Germanic languages that we discuss.

2 Background: derived *r*-verbs in Germanic

2.1 Secondary verbs

Our study concerns itself with secondary verb formation, specifically the development of a new class of secondary verbs in Germanic. Secondary, i.e., stem-derived *r*-verbs include those built to nominal, adjectival, and adverbial (and other indeclinable) bases ending in *-r*, as well as those forms where *-r-* is a morphological element and productive suffix of its own. The basic structure of verbal forms of the latter kind is root + suffix + ending. For example, the verb *folg-er-n* 'infer, deduce' in (1c-ii), consists of the basic lexical semantics of the root *folg-*, the suffixal morpheme *-er-*, and the infinitive ending *-n*. These verbs are usually classified as "deverbal" in the Germanicist handbooks because they are based on a segmentable verbal root or stem, and we follow this tradition here and also classify them as secondary.

In the following, we therefore refer to verbs derived from nominal stems as *denominal verbs*, to verbs derived from adjectival stems as *deadjectival verbs*, and to verbs derived from verbal roots or stems as *deverbal verbs*, in accordance with much of the literature. Especially in the latter two cases, there are some non-trivial diachronic and synchronic segmentation issues involved that we will discuss where relevant.²

In early Germanic, secondary verbs fall into different inflectional verbal classes based on their type of stem formation and (partly) semantics. Early Germanic continues four weak verbal classes which are relevant to secondary verb formation; however, not all of these are equally productive in all relevant older Germanic languages. In the following, we briefly discuss the relevant inflectional patterns in the four languages that are the focus of our study.

2.2 Gothic

The only documented East Germanic language, Gothic (attested ca. 4th - 9th c. CE), contains only transparently denominal and deadjectival *r*-verbs such as those in ex. (3), none of which show any iterative semantics. That is, it lacks class (1c) entirely (Wilmanns 1899: 92).

(3) Gothic secondary *r*-verbs

- a. *gaur-s* (nom.sg.m.) 'sad' → *gaur-jan* 'to sadden, make sad'

² For a detailed discussion of the morphosemantics of *deadjectival* and *deverbal* verbs and the surrounding terminology with a particular focus on the older Indo-European languages, see Grestenberger et al. 2026.

b. *maurpr* ‘murder’ (n.) → *maurpr-jan* ‘to murder’

Gothic *r*-verbs belong to the 1st weak inflectional class, or *jan*-class (*-jan*, stem-forming suffix + infinitival ending). Many members of this class are factitives, e.g., (3a), or express causativity (Garcia Garcia 2005), which has been noted to often be associated with iterative semantics (Kölligan 2007; Aikhenvald 2011; Bozzone 2020). However, we do not believe that there is sufficient evidence for a connection between the iterativity of *r*-verbs in North-West-Germanic and the causative *-jan* suffix used to form *r*-verbs in Gothic. The *jan*-causatives denote a general increased transitivity derivation independently of whether the base is nominal, adjectival, or verbal, e.g., of the type Go. *ga-drausjan* ‘cause somebody to fall’ to *driusan* ‘fall’. Moreover, as we discuss in the next section, the OHG iterative *r*-verbs almost exclusively inflect according to the 2nd weak class, so both from a semantic and from a morphological point of view, we see no plausible relation between Gothic denominal *jan*-verbs as in (3) and the type of iterativity we find in North-West-Germanic *r*-verbs.

2.3 Old High German (OHG)

Old High German (8th – 11th c. CE) *r*-verbs inflect according to the 1st, 2nd and 3rd weak classes (ending in *-e-n*, *-ō-n* and *-ē-n* respectively; *-n* again is the infinitival ending). The verbs of these classes have different semantics depending on the base and the inflectional class: Denominal verbs tend to form transitive accomplishment verbs, activity verbs, or causatives as in (4a), all of which usually follow the first or second weak class. Deadjectival verbs are typically fientives, (4b-i), factitives, (4b-ii), or statives, (4b-iii). For fientives, no clear generalization with respect to inflectional class emerges, though there is a slight preference for the third weak class (cf. ex. (4b-i)).³ The factitive class usually inflects according to the first weak class like its Gothic equivalent and the stative-intransitive class according to the 2nd or 3rd weak class. Occasional pairs like *bittar-en* ‘to make bitter, embitter’ (1st) vs. *bittar-ōn* ‘to be bitter’ (2nd), both from *bittar* ‘bitter’ show the original functional distribution most clearly, though it is increasingly obscured towards the end of the OHG period by the gradual collapse of the original inflectional class system on the way to Middle High German due to various sound changes (Wilmanns 1897: 329ff.; Paul 1904: 77), and there are only a few such “minimal pairs” in the first place. More commonly, variation with respect to inflectional class is due to dialectal or idiolectal variation.

Finally, the deverbal iterative class that is the focus of our study inflects according to the 2nd weak class, with very few exceptions.

(4) OHG secondary *r*-verbs

- a. denominal (1st & 2nd weak class)
 - i. *wuntar* ‘wonder’ → *wuntar-ōn* ‘to make wonder’ (2nd)
 - ii. *zimbar* ‘timber’ → *(gi)zimb(a)r-en* ‘to build’ (1st)
- b. deadj. fientives & factitives (1st, 2nd, 3rd weak class)
 - i. *finstar* ‘dark’ → *finstar-ēn* ‘to turn dark’ (itr.; 3rd)
 - ii. *muntar* ‘awake’ → *munt(a)r-en* ‘to wake up’ (tr.; 1st)
 - iii. *bittar* ‘bitter’ → *bittar-ōn* ‘to be angry, bitter’ (2nd)
- c. deverbal iteratives (2nd weak class)
 - i. *fliog-an* ‘fly’ → *flog-ar-ōn* ‘to fly around, flutter’
 - ii. *gang-an* ‘go’ → *gang-ar-ōn* ‘to walk around, wander’

Basically, Old High German has the same three broad classes of secondary *r*-verbs (denominal, deadjectival, deverbal) as Present Day German (cf. (1)), with the exception of the denominal class based on plural forms in *-er*, which only developed and became productive in Middle High German (MHG) (Wilmanns 1899: 93).

³ The use of this class to form denominatives is an innovation of Old High German (Wilmanns 1906: 89).

2.4 Old English (OE)

Old English (7th - 11th c. CE) also has denominal, deadjectival, and deverbal *r*-verbs, illustrated in (5).

(5) OE secondary *r*-verbs

a. denominal

i. *myrþra* 'murder' → *myrþr-ian* 'to murder' (1st)

ii. *þunor* 'thunder' → *þunr-ian* 'to thunder' (2nd)

b. deadjectival fientives & factitives

i. *mæger* 'meager' → *mæger-ian* 'to make meager' (2nd)

ii. *þeostor* 'dark' → *þeostr-ian* 'to become/make dark' (2nd)

c. deverbal iteratives

i. *flot-ian* 'swim, float' → *flot-er-ian* 'to float around; flutter' (2nd)

OE secondary *r*-verbs (and denominal and deadjectival verbs in general) inflect according to the first or second weak class, but with a different distribution than in OHG. In particular, denominal and deadjectival verbs in general tend to belong to the second weak class, like the deverbal iteratives (Brunner 1965: 330).

2.5 Old Norse (ON)

Old Norse (8th - 13th c. CE) has the same three basic classes of secondary *r*-verbs as OHG and OE:

(6) ON secondary *r*-verbs

a. denominal

i. *slātr* 'meat from a slaughtered animal' → *slātr-a* 'to slaughter'

ii. *glutr* 'waste' → *glutr-a* 'to waste'

b. deadjectival fientives & factitives

i. *fagr* 'beautiful, fair' → *fegr-a* 'to make beautiful'

ii. *snotr* 'wise' → *snotr-a* 'to make wise'

c. deverbal iteratives

i. *stak-a* 'to push, stagger' → *stak-r-a* 'to stagger'

ii. *tend-a* (1x) 'to set alight, inflame' → *tend-r-a* 'to kindle, play with flame'

Like in OE, ON secondary verbs (including the ones in *-r*) mostly inflect according to the second weak class, though among the denominal and deadjectival verbs in particular, weak class I is also found (e.g., Gutenbrunner 1951: 140; Nedoma 2001: 97-99).

2.6 Summary: derived *r*-verbs in Germanic

While East Germanic (i.e., Gothic) has only 1st weak class *jan*-verbs from nominal and adjectival bases in *-r*-, the oldest North-West Germanic languages (OHG, OE, ON) have denominal, deadjectival, and deverbal (including root-derived) *r*-verbs, with Old High German apparently preserving a functional distribution of inflectional classes with different types of secondary verbs.

An astute reader might wonder whether the derivational directionality we posit for the deadjectival and denominal verb classes described in this section is not the opposite – that is, whether we are dealing with nouns and adjectives formed from verbs through conversion or “zero derivation” (e.g., OE *myrþra* 'murder' ← *myrþr-ian* 'to murder', *þunor* 'thunder' ← *þunr-ian* 'to thunder', deverbal result nouns, instead of *myrþra* → *myrþr-ian*, *þunor* → *þunr-ian*, denominal activity verbs assumed in (5a)). We return to this issue throughout this article and in particular in Section 5, but to summarize briefly, we take the direction noun/adjective → denominal/deadjectival verb as given for these cases because 1) the morphology is additive, in that verbalizing morphology is added to the respective adjectival and nominal stems, mirroring the derivational semantics, 2) the semantics of these verb

classes closely match those of denominal and deadjectival verbs cross-linguistically (again, see especially Section 5), and 3) diachronically, suffixal *-r-* can be reconstructed for the nominal and the adjectival domain with relatively discrete functions, but not for the verbal domain. Hence there is good reason to assume that *verbal* suffixal *-r-* is an inner-Germanic development,⁴ which is also suggested by the fact that Gothic, the earliest attested Germanic language, does not have this type (Section 2.2).

With this descriptive backdrop in mind, we now turn to a more detailed investigation of the secondary *r*-verbs in the four older Germanic languages under study. Our goal is to provide a more detailed survey of the properties of their derivational bases, the distribution of the three main types, and their connection with inflectional and semantic (i.e., Aktionsart) properties in order to establish how the “deverbal” type, and hence verbalizing *-(V)r-*, developed within Germanic. In the next section, we briefly discuss the methods and sources we used to build our data set, while Section 4 provides the discussion of results.

3. Methods

The first step in our study was to create a comprehensive database of all relevant *r*-verbs in the four oldest attested Germanic languages. We relied on historical and etymological dictionaries such as Kroonen (2013; Proto-Germanic); Project Wulfila (Gothic); EWA, Bergmann (1991), Pfeifer et al. (1993) for OHG; Bosworth & Toller (1898/1921), DOE (OE), and de Vries (1977); ONP, LP (ON), as well as text archives with complex search functions (e.g., TITUS, <https://titus.uni-frankfurt.de/>; for Old High German, Referenzkorpus Altdeutsch (ReA), see <https://www.deutschdiachrondigital.de/rea/>), and, to a lesser extent, the annotated corpora available via these archives. The Old High German data were further cross-checked through a compilation of OHG *r*-adjectives and their associated verb formations based on Heidermanns (1993) and the EWA.

Due to the nature of the research question, we extracted our data via string searches using search terms such as *-ran-*, *-rian* (OE) and *-arōn-*, *-iren* (OHG). For this first phase of our study, we focused on Gothic, Old High German, Old English, and Old Norse.⁵

The main criterion for exclusion concerned *primary* verbs from roots ending in *-r-*, that is, verbs containing an *-r-* before the endings that is demonstrably part of the root rather than of a derivational/stem-forming suffix. This means we excluded forms such as, e.g., Go., OHG, OE *faran*, ON *fara* ‘move, drive’ < PIE 1. **per* ‘cross’ (LIV²: 472), OHG *dorrēn* ‘become dry’, OE *pyrrian* ‘make/become dry’ < PIE **ters* ‘become dry, dry out’ (LIV²: 637), etc., in which *-r-* is both synchronically and

⁴ See further Grestenberger & Kastner (2022) with refs. on diagnostics for derivational directionality and, e.g., Don (2005); Dahl & Fábregas (2018); Iordăchioaia et al. (2020); Iordăchioaia & Melloni (2023a,b) on directionality and the problem of ‘zero morphemes’ in conversion. Crucially, we do not think that examples such as (5a-b) can be treated as conversion no matter the derivational directionality, since the older Germanic languages had distinct noun- and verb-class marking morphology in the form of (suffixes and) declension and inflection classes at that time, no matter how semantically “bleached” these may already have been (and inherited zero-marked “root nouns” and “root verbs” had been all but eliminated). Of course, this changed later on as phonological changes eroded stem-final vowels, but at the stage under discussion (Proto-NW-Germanic and the earliest attested Germanic languages), there was no morphophonological basis for speakers to establish conversion as a derivational pattern.

⁵ Preliminary work on Old Saxon (attested 8th – 12th c. CE) and Old Frisian (ca. 13th - 16th c. CE), two additional West Germanic languages, was also conducted; so far this has yielded mostly denominal and deadjectival data (e.g., OS *mārian* ‘to proclaim’ from *māri* ‘famous’), though this is not representative due to the small amount of data sampled. We also began a preliminary investigation of Middle High German *r*-verbs in order to trace the diachronic development of the deverbal class from Old High German all the way to Modern High German. However, due to time constraints, the nature of the philological work that would be required, the accessibility (or lack thereof) of the relevant resources, and reasons of space, we ultimately decided not to include these languages in this article, though we hope to do so in future follow-up work.

etymologically root-final. Thus, given that the *-r-* in these cases was *lexically* associated with roots of various semantic types, we do not expect these to be significant for the development of the specifically iterative semantics of the deverbal *r*-suffix in North-West Germanic. The same is true for onomatopoetic verbs like OHG *zwizerōn*, *zwizzirōn* ‘twitter’ or OE *hwisprian* ‘whisper, murmur’, which cannot be considered secondary verbs in the relevant sense.⁶ In total, these criteria led to the exclusion of 32 verbs across all four Germanic languages.

All other verbs containing an *-r-* before the ending, whether from a base noun or as part of a suffix, were collected in the database, e.g.,

- OE *beterian* ‘make better, improve’ ← *betre* ‘better’ - this type is referred to as *base-r* in the following
- OE *floterian* ‘to float back and forth, flutter’ ← *flotian* ‘swim, float’ - this type is referred to as *base no-r* in the following

We further categorized and detailed all *-r-* verbs in spreadsheet entries for Meaning, Base, Base-*r* (x/- for ‘yes/no’), Base-meaning, Base-cat (i.e. was the base a noun, adjective, verb or other). If the category of the base was a noun (base-cat = n), an additional value for Base-gender was entered, though the gender of the base ultimately proved irrelevant to the research question. Where possible, additional entries were made for Aktionsart, transitivity, cognates, (earliest) attestation, place of attestation, the existence of parallel iterative or “diminutive” (*e*)/-verbs (e.g., OE *stamrian* ‘to stammer’ vs. PDG *stammeln* ‘to stammer’, cf. fn. 1), and a *notes* entry for any other useful information. The resulting verb lists for all four languages up to the *base-cat* entry can be found in the Appendix.

As mentioned above, Aktionsart was categorized using Vendler’s system (i.e., state, activity, achievement, accomplishment). The preliminary classification was made based on the verb’s meaning as entered in the dictionaries; to determine with more accuracy whether a certain verb’s semantics were, e.g., more punctual or durative, we would need to analyze the actual attestations in their context, an undertaking too large for the scope of this study (though again, we trust the previous philological work to be essentially correct in this regard). Moreover, many of the collected verbs such as OE *a-sūrian* have both stative (‘to be sour’) and factitive (durative-telic or achievement; ‘to turn sour’) semantics. In such cases, we included both meanings under a single entry if they occur with the same inflectional class, but we treated different meanings as separate entries if they also use different inflectional classes (i.e., both Aktionsart uses of *a-sūrian* are listed under the same entry, while OHG *bittar-en* ‘to make bitter, embitter’ and *bittar-ōn* ‘to be bitter, angry’ are separate entries). We believe that this best reflects the possible synchronic analyses that would have been available to speakers at the time (which is what we are interested in here). Prefixes and preverbs, whose exact semantics and productivity at the language stage in question is not yet clear, were not considered systematically at this stage of our research. In those cases where variation in this regard exists, the variants are listed under the same entry, e.g., OHG (*ir-*)*munt(a)ren* ‘to awaken’ (*muntar* ‘awake, alert’) for both the simplex and the prefixed variant. Given the overall relatively small size of entries (< 100 verbs for each language), these decisions naturally have consequences for the categorization of the data and the conclusions drawn from them. However, pending a closer analysis of the interaction of Aktionsart and inflection class change with prefixation in these classes, we think this categorization represents a valid and reliable intermediate step.

⁶ Though as an anonymous reviewer has pointed out, onomatopoetic verbs can behave like iteratives as well, particularly sound emission verbs like ‘to twitter’, ‘to murmur’, etc. Weidhaas & Schmid (2015) also discuss this phenomenon with respect to onomatopoetic (*e*)/-verbs (cf. fn. 1). We still excluded onomatopoetic *r*-verbs here because we think that it is more likely that these verbs received stem-forming morphology specifically associated with iterative morphology because of their *lexical* meaning rather than giving rise to it in the first place (a direction also discussed in Weidhaas & Schmid 2015).

4. Results

4.1 Results: base-r (y/n) in Go., OHG, OE, ON

In this section, we compare the distributional results for *r*-verbs in our Gothic, Old High German, Old English, and Old Norse corpora. We begin with Gothic, where the number of relevant *r*-verbs is limited: All of the 14 relevant derived verbs with *-r*- as part of the stem-final sequence are transparently formed from a noun or adjective with stem-final *-r* (see the Appendix for examples). In other words, there are no deverbal *r*-iteratives in Gothic.

Things are different in North-West Germanic. The results for Old High German, Old English, and Old Norse are summarized in the following figures. We begin with all derived verbs that exhibit an *-r*- in their stem-final sequence and categorize them into whether the *-r*- is part of the derivational base (base-r), not part of the derivational base (base no-r), whether it is ambiguous (that is, there is both a potential base with *-r*- and a base without *-r*- for the derived verb; examples will be given below the charts) or whether the base is unclear (the verb is derived judging from its morphological structure, but we do not know what the base is because it is unattested and/or not straightforwardly reconstructable). For example, OHG (-)*spirdaren* ‘to toil, struggle; stretch; support’ clearly has suffixal *-r*-, but no amount of digging in etymological dictionaries produced a viable synchronic or diachronic base.

In Old High German (Fig. 1), three quarters of tokens (n = 90) are verbs derived from a base stem that ends in *-r*- (67 verbs; 75%), while verbs derived from bases that do not end in *-r*- (base no-r) make up less than a quarter of tokens (10 forms; 11%), the same as forms which are ambiguous between the two classes (10 verbs; 11%). Unclear forms (3 forms, 3%) are marginal.

The ambiguous forms are the ones that are potentially interesting to our research question because these are precisely the ones in which *-r*- is ambiguous between being part of the base and being a verbalizer. For example, the aforementioned verb *wacharōn* ‘to be awake, be vigilant’ is ambiguous between being derived from the adjective *wachar* ‘awake, vigilant’ (in which case it would be “base-r, base=a”) and being derived from the same root as the 3rd weak class verb stem *wahhēn* ‘to be awake, to wake up’ (in which case it would be “base no-r, base=v”). For these “ambiguous” verbs, we recorded both possibilities in the relevant entries.

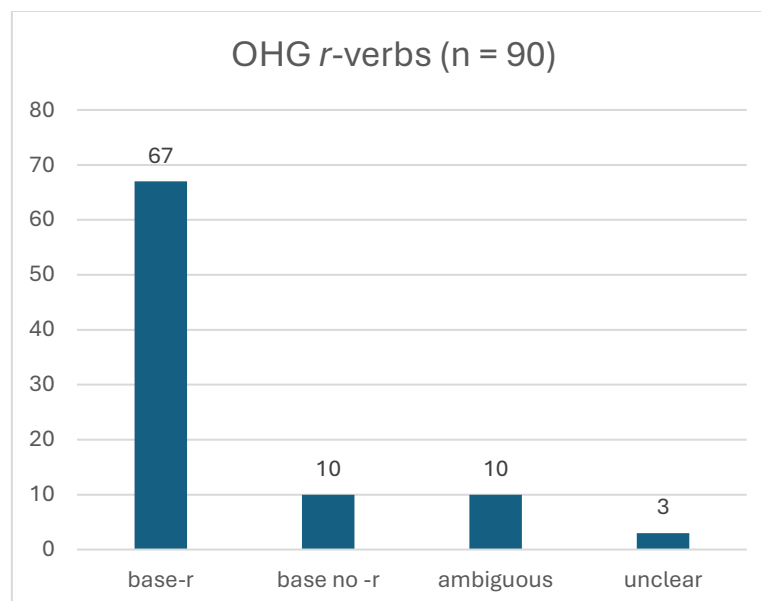


Fig. 1. OHG r-verbs

In Old English, the attested forms show a similar skew towards bases in *-r* as in Old High German (see Fig. 2): The majority of Old English tokens ($n = 69$) consists of formations in which the *-r-* is part of the base (55 verbs; 80%), while only 10 verbs (15%) are from an *r*-less base. Ambiguous (3 verbs; 4%) and unclear (1 verb, 1%) forms are marginal.

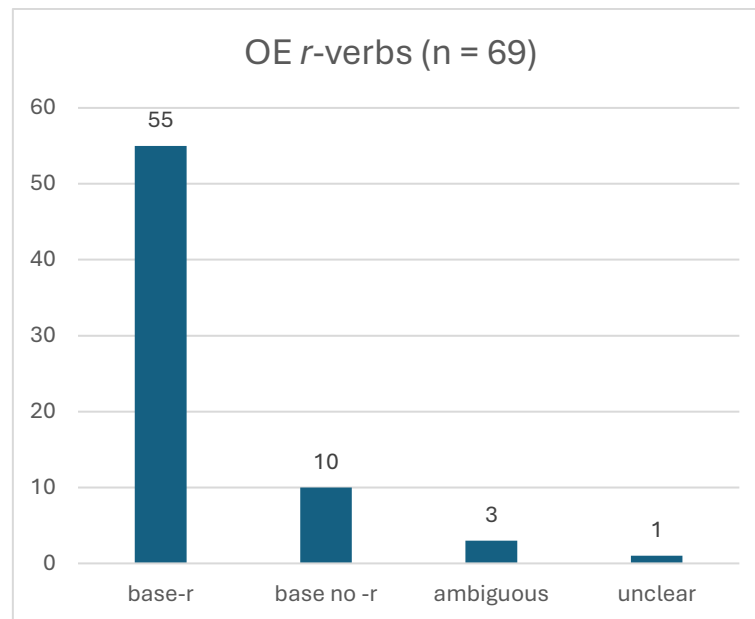


Fig. 2. OE r-verbs

In Old Norse (Fig. 3), on the other hand, only half of all tokens ($n = 80$) are verbs derived from a base in *-r* (41 verbs, 51%), which is a significantly smaller amount than in Old High German and Old English. Conversely, the number of tokens that are not derived from a base containing *-r-* is noticeably higher (32 verbs, 40%) than in Old High German and Old English. The remaining forms are ambiguous as to whether they are derived from a base in *-r-* or not (6 verbs, 8%) or derivationally unclear (1 verb, 1%).

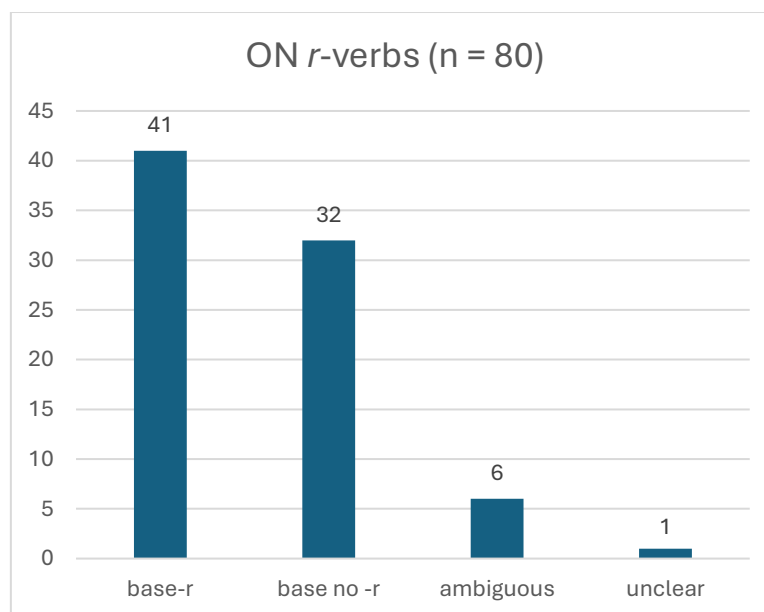


Fig. 3. ON r-verbs

To summarize, the majority of all derived *r*-verbs in Old High German, Old English, and Old Norse are synchronically based on stems containing *-r* as part of their stem-final sequence. In the next section, we take a closer look at these *base-r* verbs to determine the distribution of categories of the bases.

4.2 Results: a closer look at base-r

As before, we begin with Gothic (Fig. 4), where we find an almost equal distribution of tokens ($n = 14$) with nominal bases (8 verbs, 57%) and adjectival bases (6 verbs, 43%).

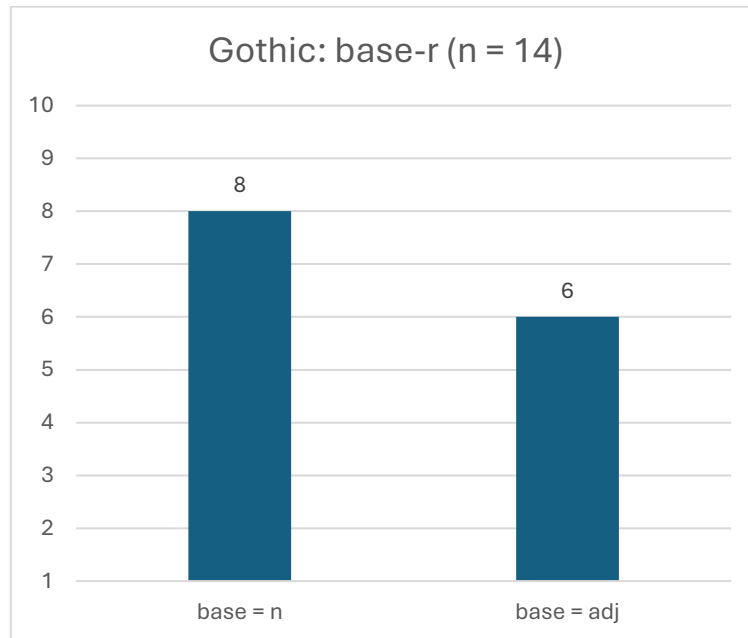


Fig. 4. Gothic base-r verbs

In Old High German (Fig. 5), the distribution is similar, with 30 verbs (45%) based on nouns and only a slightly smaller amount on adjectives (25 verbs; 37%), from a total of 67 verbs. Furthermore, four of the verbs derived from bases in *-r-* (6%) have a comparative in *-r-* as their base, while four other verbs (6%) are based on an adverb, preposition, or other indeclinable item (e.g., *widar* ‘against, opposing, opposite’ → *widerōn*, *widarōn*, *widorōn* ‘reject, oppose, compete against’). Another four forms (6%) are ambiguous. In this case, “ambiguous” means that there is more than one potential base that contains *-r-* as part of the stem-final sequence, for example, a noun and an adjective.

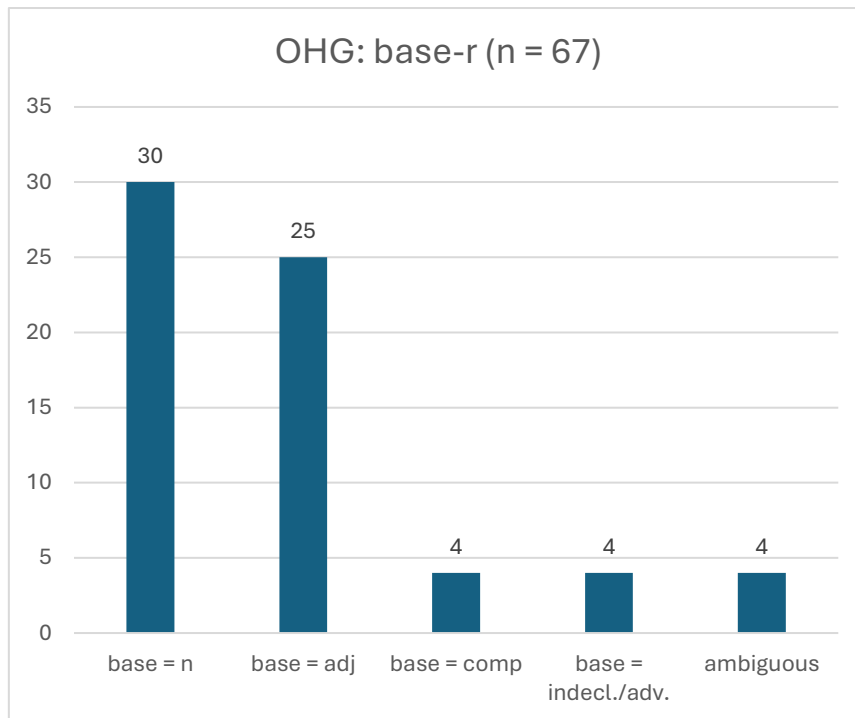


Fig. 5. OHG base-r verbs

In Old English (Fig. 6), we find the same basic types of bases, but with a slightly different distribution. About half of all tokens ($n = 55$) are from nominal bases (27 verbs; 49%), while approximately one third is deadjectival (19 verbs; 34.5%). Six verbs (11%) have adverbs or other indeclinables as bases, and there is only one verb based on a synchronic comparative (2%). Two verbs (3.5%) were classified as ambiguous with respect to their derivational base.

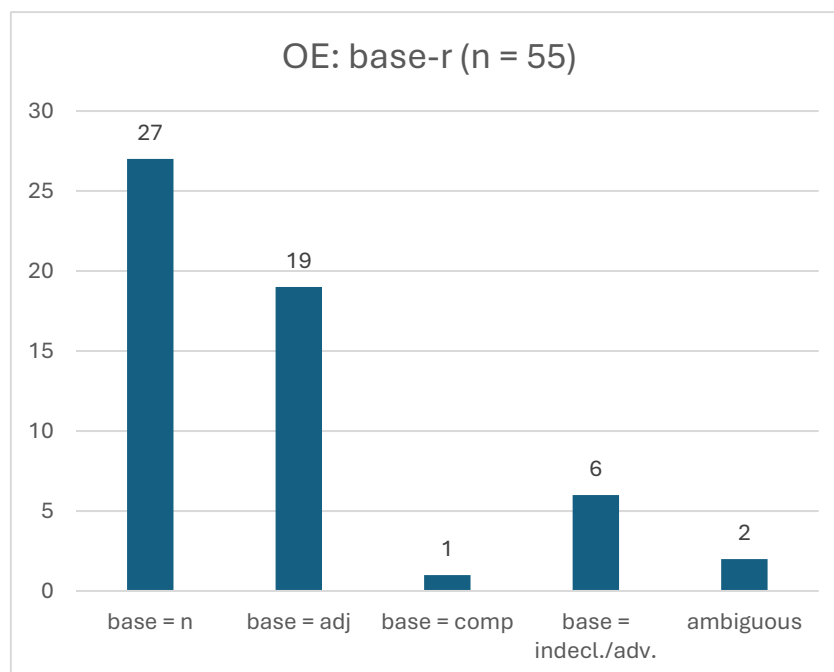


Fig. 6. OE base-r verbs

In Old Norse (Fig. 7), half of all base-*r* verbs ($n = 41$) are forms with a nominal base (21 verbs; 51%). Deadjectival verbs come in second with about one third of forms (14 verbs; 34%). Adverbial and other indeclinables are marginal as bases (4 verbs; 10%). Two verbs (5%) are derivationally ambiguous.

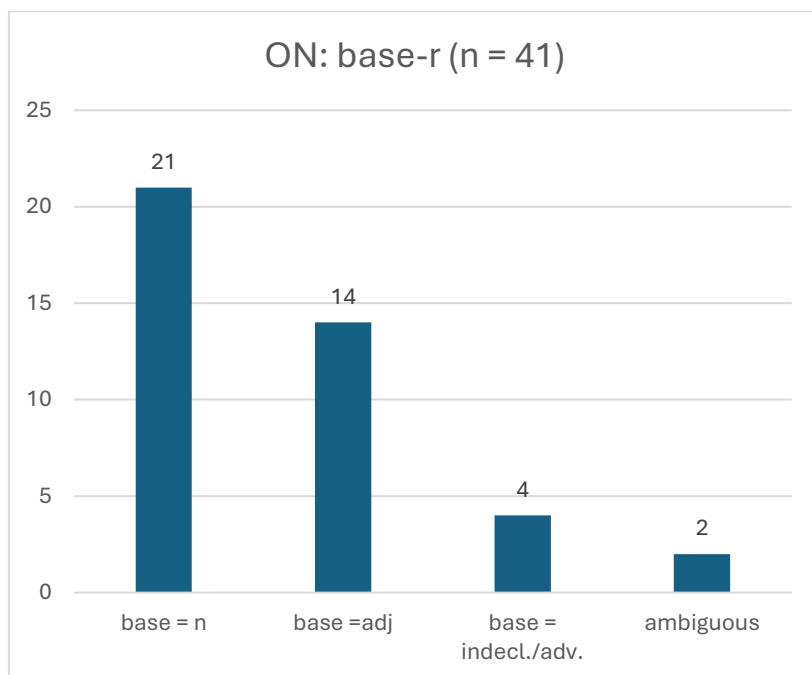


Fig. 7. ON base-*r* verbs

As an interim result, we see that base-*r* verbs are predominantly denominal and deadjectival in early North-West Germanic and exclusively so in Gothic. Comparatives in *-r-* as bases are rare in NW-Germanic (none were found in ON); we will come back to their role in deadjectival verb formation in Section 5.3. Note that the comparative marker in *-r-* is due to a Proto-NW-Germanic sound change by which *-s-* became voiced and then rhotacized in certain positions, which did not take place in East Germanic. The Gothic equivalents of NW-Germanic verbs based on comparatives therefore do not contain *-r-* but the older, non-rhotacized comparative suffix *-iz-/-oz-*, e.g., Go. comp. *minniza* ‘smaller, lesser’ → *minz-na-n* ‘to diminish, lessen’ vs. OHG *minnira* ‘smaller, lesser’ → *minner-ōn* ‘to diminish, lessen’ (note that the verbs follow different conjugational classes). Since Gothic also lacks deverbal iteratives in *-r-*, one could suspect that this is causally related to the lack of *r*-comparatives in this language. However, in Section 5.3 we argue that this explanation is less straightforward than it seems. First, we take a closer look at the secondary *r*-verbs whose base does not contain *-r-* in order to categorize their bases for Part of Speech (PoS)/category (cat).

4.3 Results: a closer look at base no *-r*

Recall that this context is not found in Gothic, where all *r*-verbs are transparently derived from nouns or adjectives in which *-r-* is part of the stem-forming suffix. The relevant languages for this context are therefore Old High German, Old English, and Old Norse. Note also that we are now dealing with very small token numbers, but we still give figures and percentages to be consistent with the previous sections.

In Old High German (Fig. 8), 90% (9 verbs) of all relevant tokens ($n = 10$) are deverbal, while only one verb (10%) is denominal.

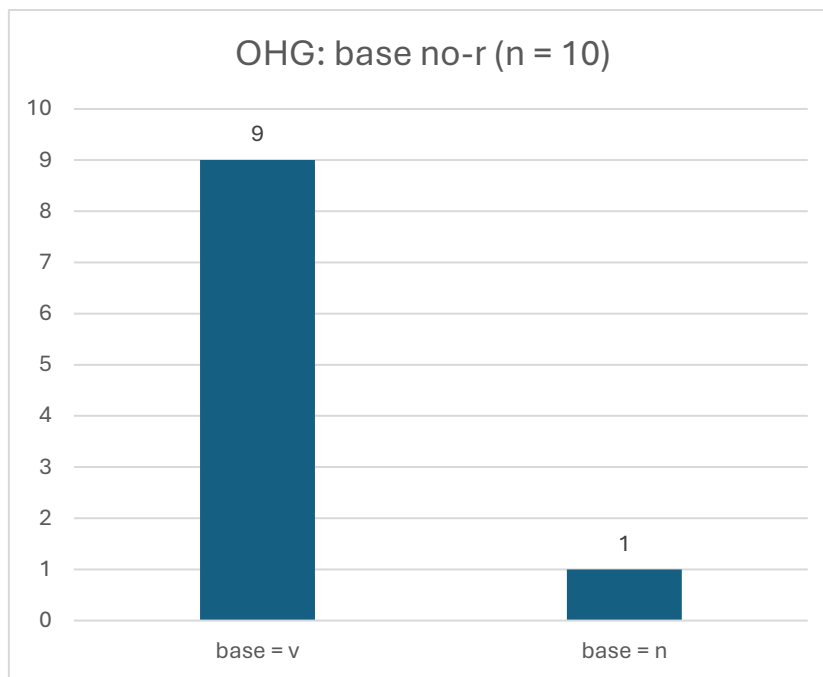


Fig. 8. OHG base no-r verbs

In Old English (Fig. 9), seven verbs (70%) of the relevant tokens (n = 10) are deverbal, while three verbs (30%) are ambiguous between denominal and deverbal or between deadjectival and deverbal formations.

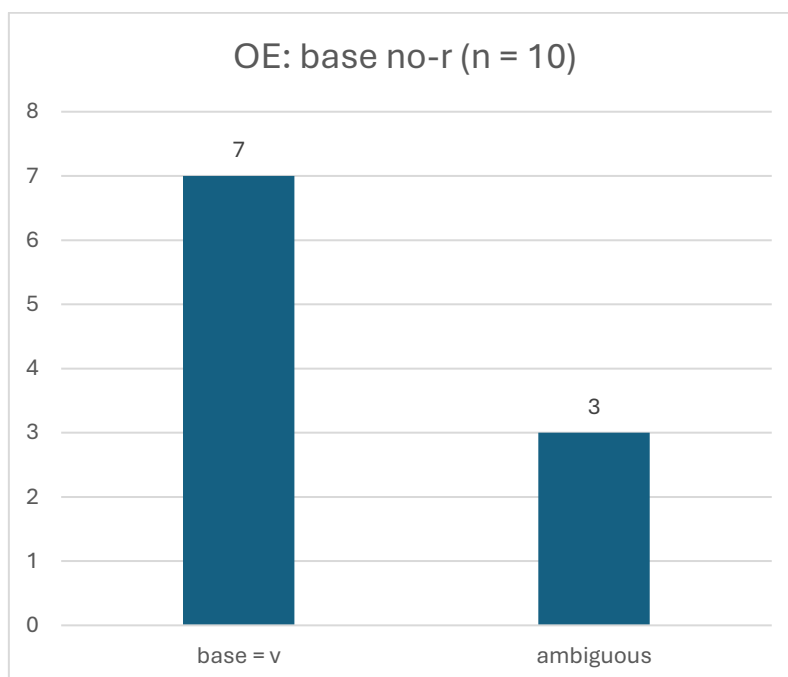


Fig. 9. OE base no-r verbs

Finally, in Old Norse (Fig. 10), most of the relevant tokens (n = 32) are deverbal (23 verbs; 72%), while two verbs (6%) are denominal. Six verbs (19%) are ambiguous with respect to their base, while the base of the remaining verb (3%) is unclear.

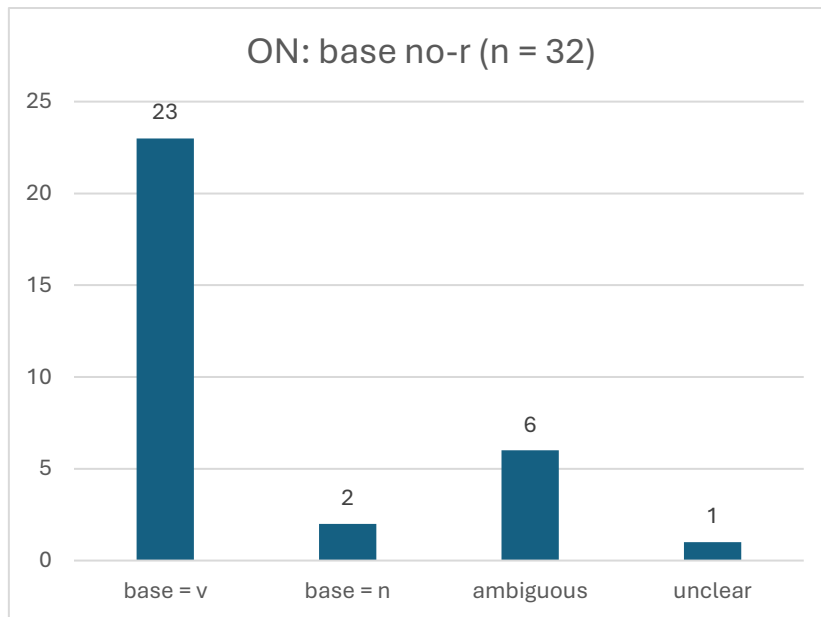


Fig. 10. ON base no-r verbs

To summarize the basic distributional insights so far, we can say that in general, if *-r-* is part of the base, the derived verb is most likely a denominal or deadjectival verb, whereas when *-r-* is not part of the base (and hence part of the verb-forming suffix), the resulting derived verb is most likely deverbal/root-derived. In the next section, we briefly discuss the Aktionsart properties of the main classes of *r*-verbs established in this section.

4.4 Results: Aktionsart

With respect to the semantics of derived *r*-verbs in the older Germanic languages, when the base is adjectival, the corresponding derived verbs are stative ('be *x*') or degree achievements/CoS verbs (fientive 'become *x*', factitive 'make *x*'), as already mentioned above. When the base is nominal, the corresponding derived *r*-verbs are mostly activity verbs ('do *x*', e.g., OHG *last(a)r-ōn* 'to slander, talk smack': *lastar* 'offense') or accomplishments (e.g., OHG *zimb(e)r-en* 'to build': *zimbar* 'building material'). This is in line with what has been observed cross-linguistically concerning the syntactic and semantic properties of deadjectival (Koontz-Garboden 2005; 2014; Bobaljik 2012; Beavers & Koontz-Garboden 2020; Fábregas 2023; Grestenberger et al. 2026) and denominal verb formation (e.g., Harley 1999; 2005; Baeskow 2019; Bleotu 2019). Although some denominal activity verbs could be construed as iteratives in terms of Aktionsart (cf. OHG *last(a)r-ōn* above), the class where we most clearly see iterativity is the deverbal class, where the derived *r*-verbs are almost exclusively activities with more or less distinctive iterative semantics, e.g., OHG *flogar-ōn* 'fly around, flutter', *gangar-ōn* 'wander around', *swebar-ōn* 'float, swim'. We return to a more detailed analysis of this class in Section 5.4.

4.5 Summary of results

In this section, we have quantitatively analyzed the derivational bases and semantics of derived *r*-verbs in Gothic, Old High German, Old English, and Old Norse. In terms of the distribution of the derivational bases, we have seen that all *r*-verbs in Gothic are transparently denominal or deadjectival, while the three North-West Germanic languages have denominal, deadjectival, and deverbal *r*-verbs. We have also observed that there is a correlation between the Aktionsart and the category of the base in these languages: If the *-r-* is part of the base, the resulting verb is denominal or deadjectival, and semantically falls into one of the classes that are cross-linguistically expected for denominal and deadjectival verb formation (activities and accomplishments vs. degree achievements and states, respectively). These

verbs are therefore a subclass of denominal and deadjectival verb formation in these languages more generally that just happens to be based on nominal and adjectival stems in *-r-*.

The smaller class of deverbal *r*-verbs, on the other hand, consists mostly of activity verbs, and the *-r-* is not part of the base in this class, which suggests that the *-r-* is indeed an activity-forming verbalizer with iterative Aktionsart in this class. Since Gothic has no deverbal *r*-verbs, this means that the verbalizing function of *-r-* must be a common North-West Germanic innovation. In the next section, we develop an analysis of how this *-r-* found its way into the verbal domain in North-West Germanic. However, we want to emphasize once again that we are dealing with a very small number of tokens for this class: Like in the modern Germanic languages, the verbal *-r*-suffix was never a hugely productive strategy for expressing iterative Aktionsart. Nevertheless, it is a discernible class whose diachrony should be accounted for.

5. Analysis: getting adjectival morphology into the *v*-domain

5.1 Introduction

The starting point for our analysis is that *-(e)r-* is not an inherited Proto-Germanic verbalizer because no such verbalizer can be reconstructed for Proto-Indo-European and because as we have seen in Section 4, it is unattested in East Germanic (Gothic). This suffix must therefore have become reanalyzed as a verbalizer at some point in (proto-)North-West Germanic. In this light, it is reasonable to assume that this happened in the context of cross-categorial derivation, that is, in the formation of denominal or deadjectival verbs to stems that contained *-r-*, since these contexts are amply attested in all older Germanic languages (including Gothic) and because this is a context in which diachronic *missegmentation* of stem-forming elements often leads to the development of new categorizers (Grestenberger 2022; 2023). To model this change, we use Distributed Morphology, a realizational approach to morphology in which word structure directly mirrors syntactic structure. Complex words are built in the syntax by merging an acategorial root with a categorizer – a word-class determining functional head *v* (verbalizer), *n* (nominalizer), or *a* (adjectivizer/stativizer), represented as “*x*” in Fig. 11 – and subsequently combining with “higher” functional heads such as “*y*” in Fig. 11. Abstractly, morphemes represent syntactic terminal nodes which are phonologically realized in accordance with the Subset Principle under certain locality conditions (cf., e.g., Bobaljik 2000, 2012; Bobaljik & Harley 2017; Embick 2010, 2012, 2015; Choi & Harley 2019 a.m.o.), indicated with a capital “*X*” and “*Y*” in Fig. 11 (note that these exponents can also be phonologically empty, i.e., zero). We follow Bobaljik 2012, Bobaljik & Harley 2017 in assuming a strict version of locality in which terminal nodes can only “see” structurally adjacent nodes within the same phase for purposes of phonological realization (“Vocabulary Insertion”, VI, in DM terms). That is, in a structure like Fig. 11, the phonological realization of *x* can be determined by the root or the node *y*, but not a structurally higher head *z* above *y* since in that case *y* would intervene.⁷ What is crucial for our purposes is that these locality conditions also hold diachronically, in the course of “missegmentation” or, rather, reanalysis of the exponents of terminal nodes. That is, given the structure in Fig. 11, the phonological content and syntactico-semantic features associated with *x* can become reanalyzed as associated with *y*, but not with a putative higher head *z* across the intervening *y*. This reanalysis is indicated by the arrow in Fig. 11.

⁷ Because VI is strictly “bottom-up” (e.g., Bobaljik 2000, 2012, Embick 2012, 2015), only the abstract syntactico-semantic features of *y* can determine VI of *x* in this scenario, not the phonological content of *y* since this has not been inserted at this point. See further Embick 2012 on “inward-” vs. “outward-looking” contextual allomorphy.

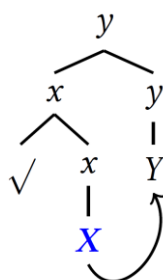


Fig. 11. “Upwards” morpheme reanalysis in a complex word

In the following, we argue that this type of “upwards” reanalysis in complex word forms is exactly what gave rise to verbal *-(V)r-* in the older Germanic languages, where “X” in Fig. 11 is to be identified with the *-(V)r-*morpheme in the context of cross-categorical derivation and “Y” with the exponent of the selecting verbalizer. We therefore need to look for a nominal or adjectival suffix in *-r-* that could have become reanalyzed as verbal in these contexts.

There is no *nominal* suffix in the older Germanic languages that would be an obvious candidate for such a reanalysis. While there are some inherited nouns that contain *-r-* as part of their stem-forming morpheme, these do not form a uniform semantic class and are not productive in any way: They include inherited kinship terms such as **fader-* ‘father’, **mōder-* ‘mother’ and the inherited words ‘fire’ and ‘water’ (Streitberg 1963: 205-207; Krahe & Meid 1969b: 77-78; Bammesberger 1990: 204-208), as well as some inherited concrete nouns such as Proto-Germanic (PGmc.) **timra-* ‘timber, wood’ and some inherited instrument nouns with the complex suffix **-pro-* (< PIE **-tro-*) such as **rōpra-* ‘rudder’ (Kluge 1926: 49-50; Krahe & Meid 1969b: 79-80). While some of the derived *r-*verbs in our corpus are undoubtedly based on concrete and instrument nouns, these are precisely the ones that are *not* associated with iterativity, as we saw in Section 4, so they are an unlikely starting point of the reanalysis.

Another nominal class that can be excluded are the nouns of profession/agent nouns in PGmc. **-ārja-* (e.g., OHG *jag-ōn* ‘to hunt’ → *jag-āri* ‘hunter’, OE *writan* ‘write’ → *writere* ‘writer’) that gave rise to the productive agent noun suffixes of the modern Germanic languages (e.g., Engl. *-er*, PDG *-er*). This suffix is generally considered to have its origin in Latin loanwords formed with the suffix *-ārius* (see Scherer 2005: 83-88 with lit.), a denominal suffix that formed relational adjectives and animate relational nouns in Latin (e.g., Lat. *molina* ‘mill’ → *molin-ārius* ‘someone who has to do with mills; miller’) and the older Germanic languages, and subsequently also deverbal agent nouns (Krahe & Meid 1969b: 81-83). This loan suffix seems to have fallen together with phonologically similar inherited Germanic suffixes such as **-warja* in some branches, e.g., OHG (Kluge 1925: 31-32; 1926: 19; Scherer 2005: 87). However, there were no *r-*verbs in our corpus for which such an animate agent(-like) noun could plausibly have been the derivational basis,⁸ and Old Norse, which was geographically and culturally farther removed from the sphere of Latin influence and in which these agent nouns took hold less quickly (Krahe & Meid 1969b: 82) still has deverbal iterative *r-*verbs. Moreover, while denominal verbs based on such agent(ive) nouns do tend to form activity verbs meaning ‘act like x’ or ‘be (like) x’, these are not usually associated specifically with iterativity (Bleotu 2019; Marescotti &

⁸ We only found one potential case, OHG *lōwirōn* ‘to tan’, as attested in the preterit participle *gilouroth*: ‘tanned; *sine lana*’ in a 12th c. Alemannic gloss, besides which a denominal noun of profession **lōwāri-* ‘tanner’ is posited by EWA V, 1480 (<https://ewa.saw-leipzig.de/articles/l%C3%B4wari/>) based on attested MHG *lōwere*. However, since the noun of profession is attested later than the participle and the participle’s meaning suggests a verb ‘apply *lō* (tanbark) to sth.’, we have classified this as a denominal verb to OHG *lō* ‘tanbark’, as suggested by EWA (loc. cit.). Moreover, we would expect a verb in *-arōn* rather than *-irōn* if this were from the noun of profession.

Grestenberger forthcoming). We can therefore also exclude these nominals as the potential diachronic base of our deverbal *r*-verbs.

This leaves deadjectival verbs as the starting point for the reanalysis. In the next sections, we provide a survey of the two major classes of *r*-adjectives in Germanic and their role in the development of the iterative verbal class.

5.2 Types of *r*-adjectives in Germanic

Germanic inherited several variants and derivatives of an adjective-forming suffix containing *-r-*: PGmc. **-ra-* (< PIE **-ro-*), **-ri-* (PIE **-ri-*), and the inner-Germanic innovation **-r-ja-*. In general, these formed two different types of adjectives in Germanic (most of the examples are formed with **-ra-*, the most productive allomorph): Type 1 consists of root-derived Property Concept adjectives, i.e., adjectives from stative, prototypically adjectival roots (Dixon 1982; Thompson 1989; McNally & Boleda 2004; Alfieri 2014; Francez & Koontz-Garboden 2015; 2017) such as Engl. *great, wild, tall, red, broad, wide*, etc., whereas type 2 consists of root-derived “verbal adjectives” from eventive, “verbal” roots, overwhelmingly ones that form strong verbs in the older Germanic languages. Reconstructable examples for type 1 are given in (7) and for type 2 in (8), (Wilmanns 1899: 430-431; Krahe & Meid 1969b: 78-81; Heidermanns 1993).

- (7) Proto-Germanic PC adjectives
 - *dap-ra-* ‘heavy’
 - *haid-ra-* ‘clear’
 - *hai-ra-* ‘grey’
 - *mag-ra-* ‘meager’
- (8) Proto-Germanic verbal adjectives
 - *bit-ra-* ‘biting, bitter’ : *beit-a-* ‘to bite’
 - *klib-ra-* ‘sticky’ : *kleib-a-* ‘to stick’
 - *slip-ra-* ‘slippery’ : **sleip-a-* ‘to slip’
 - *wak-ra-* ‘awake, aware’ : *wak(na)-* ‘to wake up’
 - *wit-ri-* ‘knowing, knowledgeable’ : **wait-* ‘to know’
 - *skei-ri-* ‘shining’ : **skei-* ‘to shine’

If one of these classes (or both) was the locus of reanalysis of deadjectival verbs as iteratives, the compositional semantics of the verbs derived from such adjectives should provide some semantic overlap or ambiguity that could have led to such a reanalysis. We therefore have to take a closer look at the morphosemantics of deadjectival verbs derived from such adjectives.

5.3 Morphosemantics I: Deadjectival verbs & degree achievements

Verbs derived from gradable adjectives of type 1 (PC adjectives) form degree achievement verbs, that is, CoS verbs of the factitive-fientive alternation (‘make *x*’ - ‘become *x*’), as well as statives (‘be *x*’). The properties of these verbs have received much attention in the theoretical literature (e.g., Hay et al. 1999; Koontz-Garboden 2005; 2014; Kennedy & Levin 2008; Bobaljik 2012; Beavers & Koontz-Garboden 2020; Fábregas 2023; Gast et al. 2025; Vyshnevskaya 2025a,b) and are relatively well understood: The underlying patient of these verbs undergoes a change of state between the degree to which a property *P* (e.g., *wide*) holds at the beginning of the event and the degree to which it holds at the end. These CoS verbs can be transitive, with an external argument that causes the change of state, or intransitive. Depending on the adjective, different scales of measurement for these degrees may be relevant.

(9) Degree achievements

- a. *They widened the road* (factitive)
b. *The road widened* (fientive)

If we turn now to the deadjectival *r*-verbs in the older Germanic languages in our corpus, we see that verbs derived from such PC adjectives display exactly these expected degree achievement and state semantics. Representative examples from Old High German are given in Table 2, but the same can be observed in Gothic, Old English, and Old Norse (see the Appendix). As mentioned in Section 2.3, OHG shows traces of a differentiation along inflectional class: weak class III tends to be found in the stative and fientive class, while factitives tend to use weak class I and II.

Statives & fientives	Meaning	Base	Meaning
(h)lūttar-ēn	be bright, clear	lūt(t)ar, hlūt(t)ar	bright, clear
finst(a)r-ēn	become dark	finstar, finstir, finster	dark
timber-ēn	become murky	timber	somber, murky
Factitives	Meaning	Base	Meaning
bezzar-ōn	to make better	bez(z)ero	better
bittar-en	make bitter	bittar	bitter
(gi)lūttir-en, liut(e)r-en	clear, make bright	lūt(t)ar, hlūt(t)ar	bright, clear
heitar-en	be/make cheerful	heitar	clear, cheerful
irbar-ōn	make bare	bar	bare
magar-en	make thin	magar	meager, thin
(ir-)munt(a)r-en	make alert	muntar	awake, alert
sūbir-en, sūb(e)r-en, sūbar-ōn	make clean	sūber, sūb(i)ri	clean

Table 2: Deadjectival statives and degree achievement verbs in Old High German

The use of weak class I inflection in this group of verbs also explains one of the important morphophonological properties that distinguishes the factitive class from the iterative class of secondary *r*-verbs: The PGmc. vowels **ǣ*, **ō*, **ū* underwent umlaut (i.e., vowel fronting) before **i* and **j* in the North-West Germanic varieties; albeit at slightly different points in time, with North Germanic earlier than the other varieties (e.g., Krahe & Meid 1969a: 59-60; Ringe & Taylor 2014: 222-236). Therefore denominal and deadjectival class I verbs with the relevant root vowels underwent umlaut before the class I suffix *-*j-an*. Old High German spelling does not usually reflect this, but we see it in ON cases such as *fagr* ‘beautiful’ → *fegra* ‘to beautify, adorn’, *magr* ‘meagre’ → *megra* ‘to make meagre, emaciate’, *stōrr* ‘big, great’ → *stæra* ‘to augment, increase’, etc., and in OE in, e.g., *attor* ‘poison’ → *ættrian* ‘to poison’, *sundor* ‘apart’ → *syndrian* ‘to divide, separate’. OHG class I verbs such as *swangeren* ‘to impregnate’ (from *swangar* ‘pregnant’), *snuoren* ‘to string, lace up’ (from *snuor* ‘string’) and *lūttaren* ‘to purify’ (*lūttar* ‘pure’) also underwent *i*-umlaut, as their MHG and PDG equivalents *swengern*/*schwängern*, *snüeren*/*schnüeren*, and *liutern*/*läutern* show (needless to say, the spelling of the umlauted vowels varies wildly across the Germanic varieties). The situation is complicated by the fact that there are also class I verbs with the relevant root vowels that do not show umlaut for different reasons, that there are *i*-variants of the adjectival *r*-suffixes (*-*ri*-, *-*rja*-, cf. Section 5.2) which also trigger umlaut, and that ON also has *u*-umlaut, which shows up both in the factitive and the iterative class. However, the overall generalization is that secondary class I verbs *can* display *i*-umlaut while secondary class II verbs never do. There is a second morphophonological property that sets the factitive and the iterative class apart, namely ablaut, which we discuss in Section 5.4.

Concerning the morphosyntax of the classes of verbs in Table 2, Bobaljik (2012) has argued that deadjectival degree achievements are derivationally based on the comparative rather than the positive (see also Gast et al. 2025, Vyshnevskaya 2025a,b). Part of the argument is based on morphological evidence from verbs based on suppletive adjectives; these always select the allomorph of the comparative rather than the positive, e.g., (10).

(10) Deadjectival verbs from comparatives of suppletive adjectives (cf. Bobaljik 2012: 190-192)

- a. Engl.: *to worsen* (**to bad(den)*); *to better* (**to good(en)*)
- b. PDG: *ver-besser-n* 'to better' (*gut* 'good', comp. *besser* 'better')
- c. Lat.: *melior-āre* 'to better' (*bonus* 'good', comp. *melior* 'better')

But comparative forms of non-suppletive adjectives can also form degree achievements:

(11) Deadjectival verbs from comparatives of non-suppletive adjectives (Bobaljik 2012: 175)

- a. Engl. *low* – *low-er* → *to low-er*
- b. PDG: *schön* 'beautiful' – *schön-er* 'more beautiful' → *ver-schön-er-n* 'to beautify, embellish'; *groß* 'big' – *größ-er* 'bigger' → *ver-größ-er-n*

Based on this morphological evidence and the scalar degree semantics of deadjectival verbs (i.e., the end point of the change of state must have progressed along a scale *in comparison with* the starting point), Bobaljik argues that such verbs are *always* based on the comparative, even in cases like (12) in which the comparative morpheme is not overt.

(12) Deadjectival verbs based on the positive

- a. Engl.: *cool* → *to cool* (comp. *cool-er*); *dark* → *to dark-en* (comp. *dark-er*)
- b. PDG: *sauber* 'clean' → *säuber-n* 'to clean' (comp. *sauber-er* 'cleaner'); *dunkel* 'dark' → *ver-dunkel-n* 'to darken' (comp. *dunkl-er/dünnl-er* 'darker'); *rot* 'red' → *röt-en* 'to redden, to make red' (comp. *röt-er* 'redder')

Bobaljik (2012: 206) proposes that in these cases the verbalizer itself expresses the semantics of comparison, parallel to the role of the overt comparative morpheme in deadjectival degree achievements based on suppletive adjectives. Moreover, because of the iteration of temporal stages across the relevant scale — e.g., from 40° to 17° (*the soup cooled*) or from two metres to four metres (*they widened the road*) — one could speculate that the degree achievement nature of these verbs could give rise to an iterative reading. From this perspective, it would not be surprising if these structures diachronically also gave rise to new *overt* verbalizing material expressing comparison (i.e., degree achievement verbalizers) and, subsequently, iteration, through a reanalysis of overt comparative morphology such as proto-NW-Germanic. *-(a)r- in the context of deadjectival verb formation, as in (11). This development could have been reinforced by the presence of positive adjectives in which *-(a)r- was an adjectival stem formant and in which the comparative morpheme is not contained in the derived verb, as in PDG *sauber* 'clean' → *säuber-n* 'to clean'. This diachronic reanalysis is illustrated in Figure 12 using the PDG forms as a stand-in for what would have been a proto-NW-Germanic reanalysis of comparative (and positive?) *-(a)r-. The arrows mark the structural position of the reanalyzed element before and after the change.

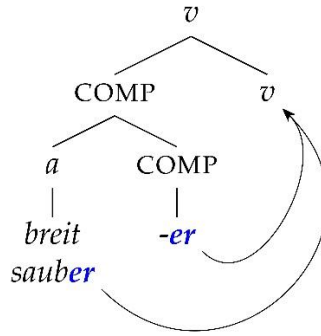


Fig. 12: reanalysis of positive/comparative *-(e)r* as verbalizer/Aktionsart marker

However, if degree achievement verbs such as the ones in Table 2 (either from morphological positives or from comparatives in *-(e)r*) were the source of such a reanalysis, we would still expect the resultant verbalizer to derive factitive-fientive degree achievements rather than iterative activity verbs, given 1) the semantics of the deadjectival verbs based on such adjectives and 2) the historically attested parallels in which this type of reanalysis has taken place. Consider, for example, the diachronic development of the PDG verbalizer *-ig-* in, e.g., *rein* ‘clean, clear’ → *rein-ig-en* ‘to clean, make clear’, *satt* ‘full, sated’ → *sätt-ig-en* ‘to satiate’, etc. Historically, this was a relational adjectival suffix that was primarily used to form adjectives from nouns, e.g., OHG *heil* ‘good fortune’ → *heil-ag*, *heil-ig* ‘with good fortune; holy’; *nōt* ‘misery, distress’ → *nōt-ag*, *nōt-ig* ‘miserable, in distress’. These adjectives could become the input to deadjectival verb formation (OHG *heil-ag-ōn*, *heil-ig-ōn* ‘make/proclaim to be holy’; *nōt-ag-ōn* ‘make miserable; force sbdy.’). In Middle High German, the adjectival suffix was increasingly interpreted as a verbalizer and used to form deadjectival factitives from adjectives that did not end in *-ig-* (MHG *-ec-*, *-eg-en*), but also increasingly from nouns (*Erde* ‘earth’ → *be-erd-ig-en* ‘cover with earth; bury’), cf. Wilmanns (1899: 111-113). The reanalysis must have taken place in contexts in which both a derived (historically denominal) and an underived (primary) adjective became the input to deadjectival verb formation, as in (13a), in which the verb based on the *-ig-* adjective could be reanalyzed as associated with the primary, underived adjective (13b).

(13) Reanalysis of adjectival *-ig* → verbal *-ig*, illustrated with MHG examples

a. <i>rein-ec</i>	‘clean’	:	<i>rein-eg-en</i>	‘to clean’
b. <i>(h)rein(i/e)</i>	‘clean’	:	<i>(h)rein-en</i>	‘to clean’
c. <i>sat</i>	‘sated, full’	:	<i>set(t)-ig-en</i>	‘make full, sated’
(replacing older <i>set(t)-en</i> , <i>sat(t)-en</i>)				

The *r*-verbs that show the iterative (rather than degree achievement) reading, on the other hand, are precisely the ones that are *not* associated with PC roots or adjectives, either synchronically or diachronically. That is, if verbs such as in (10)-(12) had been the starting point for the grammaticalization of the iterative verbalizer, for example through a conventionalization of the iteration of temporal stages inherent to degree achievements, *we would still expect to see this development in degree achievements first*, i.e., in deadjectival verbs from PC adjectives. However, as we have argued based on our corpus study, such verbs are robustly fientive-factitive alternation verbs at the stage at which the iterative verbalizer already exists and consistently occurs with eventive (i.e., non-PC) roots; they also tend to take weak class I inflection in OHG and ON (albeit not consistently), while the iterative class consistently occurs with weak class II inflection.

Thus, while we will ultimately argue that a *structurally* similar reanalysis to the one illustrated in Fig. 12 gave rise to deverbal *r*-verbs, we do not think that this happened in deadjectival degree

achievement verbs, since these contexts would have given rise to a factitive or factitive-fientive verbalizer, as the development of German *-ig-* in (13) shows, not to an activity/iterative verbalizer.⁹

Given these considerations, we are now essentially left with the second class of Proto-Germanic *r*-adjectives, the so-called verbal adjectives in ex. (8), as the starting point for the reanalysis. In the next section, we begin with a discussion of iterative verbs and then turn to the behavior of verbs derived from such verbal adjectives.

5.4 Morphosemantics II: Iteratives

In line with much of the previous literature (Dressler 1968; Cusic 1981; Wood 2007; Tovená & Kihm 2008; Tovená 2010; Greenberg 2010; Grestenberger & Kallulli 2019; Kastner 2020), we take iteratives to be event-internal pluractional verbs designating events that consist of a plurality of subevents such as Engl. *flutter*, *nibble*, *hop*, *waver*, *knock*, *jump*, etc. Depending on the language and the morphological marking they receive, these verbs are also called (or overlap with verb classes called) “intensives” or “verbal diminutives”; some of them can be described as iterated semelfactives (= activities). The exact semantic contribution of the iterative marker in such verbs is a matter of debate, and the iterativity of these verbs has even been argued to be epiphenomenal or entirely dependent on the interpretation of the base (Oltra-Massuet & Castroviejo 2014; Kastner 2020). What is important for our purposes is that iteratives/pluractionals are *atelic activity verbs*, and thus syntactically and semantically very different from telic degree achievement (CoS) verbs (though as mentioned, these can have atelic readings as well). We also follow the literature in assuming that the iterative morpheme is an event *modifier* rather than a functional head: Kastner (2020: 61–62), for example, proposes that the Hebrew intensive template combines with an event modifier *vACTION* that adjoins to the *vP* and can return a pluractional reading depending on the meaning of the embedded root.

Turning now to our Germanic iterative *r*-verbs, it is clear that these can also be characterized as event-internal pluractionals, as illustrated with the pairs of verbs in Table 3 for Old High German. Note that the meaning of the “iterative” is often identical to (or at least difficult to distinguish from) the meaning of the verbal base, especially if the base is itself an activity verb.

⁹ Further examples for the development of adjectival suffixes into factitive verbalizers are found in the history of Greek, e.g., adjectival suffix *-u-* → deadjectival verb *-ū-ne/o-* (e.g., *kratú-* ‘strong’ → *kratú-ne/o-* ‘to strengthen’); whence the factitive verbalizer *-ūne/o-* (e.g., *mégas*, *mégal-* ‘great, big’ → *megal-ūne/o-*); cf. Tucker 1981; Villanueva Svensson 2024. We know of no cases in which adjectival suffixes associated with the formation of primary (PC) adjectives give rise to iterative verbalizers.

However, there is a possible caveat with respect to the role of comparative morphemes in such a development: As Jonathan Bobaljik has reminded us, there are languages in which comparative morphology can also have attenuative or similitive meaning (cf. English *green-ish*), e.g., in some Uralic and Turkic languages (Bobaljik 2012: 25), in Germanic (Jäger & Walkden 2021), and in Baltic (Petit 2021), and this is generally interpreted as evidence for a diachronic development path *SIMILATIVE* > *COMPARATIVE* (Jäger 2010, 2018). In languages in which attenuative morphemes can also combine with verb stems (so-called “verbal diminutives”), the resulting verbal meaning can be similar to that of a pluractional, e.g., Halkomelem *lhí:m* ‘picking’ → reduplicated verbal diminutive *lhi-lhi:m* ‘picking a little bit’ (Wiltschko & Steriopo 2007: 3; cf. adjectival *p’eq’* ‘white’ → reduplicated attenuative *p’l-p’eq’* ‘a little white, whitish’); Hebrew *caxak* ‘to laugh’: *cixkek* ‘to giggle’, *kafac* ‘to jump’: *kifcec* ‘to jump around’ (De Belder et al. 2014: 157). Note that these are precisely the languages in which the respective diminutive morphemes behave like (root) modifiers (Wiltschko & Steriopo 2007) or at any rate as attaching below the first categorizing affix (De Belder et al. 2014). However, the inherited comparative morpheme that gave rise to the North-West Germanic comparative suffix, **-is-/*-jos-*, does not seem to be associated with attenuative(/similitive/elative) meaning or behave like a modifier (rather than a head) in any of the languages in which reflexes are attested, including Germanic. Therefore, although a potential diachronic path *ATTENUATIVE* > *ITERATIVE* in the verbal domain deserves further cross-linguistic investigation, we do not think that it played a role here.

Iterative		Base	
<i>bliuw-ar-ōn</i>	beat	<i>bliuwan</i>	make blue, beat
<i>flog-ar-ōn</i>	fly around, flutter	<i>fliogan</i>	fly, float, glide
<i>gang-ar-ōn</i>	walk around, wander	<i>gangan</i>	go, walk
<i>irscab-ar-ōn</i>	scrape together, snatch	<i>(ir)scaban</i>	scrape, scratch
<i>sweb-ar-ōn</i>	float, drift	<i>swebōn</i>	float, drift
<i>uob-er-ōn</i>	practise, peruse	<i>uoben</i>	practice, do
<i>wimm-er-en</i>	swarm, teem with	<i>wium(m)en, wimmen</i>	swarm, teem with

Table 3: OHG (*a*)*r*-iteratives (event-internal pluractionals)

Since we have seen in the previous section that deadjectival verbs from type 1 adjectives (PC adjectives) gave rise to degree achievement verbs, we now pursue the hypothesis that type 2 adjectives (“verbal adjectives”) were the basis of iteratives. Specifically, this predicts an association between *r*-iteratives and type 2 *r*-adjectives in our corpus. More precisely, since iteratives are usually described as deverbal in the literature and deverbal *r*-verbs are almost exclusively derived from bases that do not contain *-r* (see Section 4), we need to focus on those *r*-verbs that were classified as ambiguous between “base-*r*” and “base no-*r*” in our collection. As we saw in Section 4, the relevant class of verbs is very small, but it leaves us with a handful of verbs that are usually characterized as iterative and that are associated both with a type 2 adjective (either directly attested or reconstructable based on inner-Germanic cognates) and their base verbs (again either directly attested in the same branch or reconstructable based on inner-Germanic cognates), usually strong verbs. These pairs are summarized in Table 4. Further and more detailed etymological investigations will undoubtedly unearth more such pairings.¹⁰

<i>r</i> -iterative		type 2 adj. in <i>-r</i>		verbal base	
OHG					
<i>wahhar-ōn</i>	be alert, vigilant	<i>wahhar</i> , <i>wachar</i>	alert, vigilant	<i>wach-ēn</i>	be awake, vigilant
<i>weigar-ōn</i>	be obstinate, refuse	<i>weigar</i>	obstinate, combative	<i>wīg-an</i>	fight, oppose
<i>flogar-ōn</i> (<i>flok-</i>), ¹¹ ON <i>flōgra</i>	flutter, flicker	* <i>flakra-</i> , OE <i>flacor</i>	flickering	* <i>flakk/g-ōn</i> , ME <i>flakken</i>	flutter, flicker
<i>ir-lungar-ōn</i>	wander around	<i>lungar</i>	capable, diligent	<i>gi-lingan</i>	succeed
OE					
<i>stam(e)r-ian</i>	stammer	<i>stamor</i>	stammering	* <i>stimm-an</i> , MHG <i>stamen</i> , <i>stemmen</i>	stop, falter

¹⁰ This is especially true for some of the inherited ON *r*-iteratives such as *flaðra* ‘to stammer, flatter’ (de Vries 1977: 127-128; ONP <https://onp.ku.dk/onp/onp.php?o21672>), cf. MHG *vleder(e)n* ‘flutter’, OHG *fledar-mûs* ‘bat’, lit. ‘flutter-mouse’; see EWA s. v. ‘fledarmûs’ (<https://ewa.saw-leipzig.de/articles/fledarm%C3%BBs>). The editors explicitly reconstruct an OHG *r*-verb **fledirōn* based on an adjective or noun **fledar*, itself based on the reconstructable verbal root **fled-* attested in the OHG participle (nom. pl.m.) *fledenta* ‘fluentes’ ‘hanging loose’ in a 12th c. gloss. This derivational chain is thus entirely parallel to the pairings in Table 4, but we have excluded it here because its reconstruction presupposes the derivational chain that we are arguing for here and because of certain formal and semantic discrepancies between ON *flaðra* and reconstructed OHG **fledirōn*.

¹¹ See Kroonen (2013: 144 s. v. **flakk/gōn- 2*); de Vries (1977: 135); ÁBM s. v. *flōgra* on this equation.

<i>slidr-ian</i>	slip, slither	<i>slidor</i>	slippery	<i>slidan</i>	slide
<i>sicer-ian</i>	trickle, sink down	* <i>sigra-</i> , MHG <i>seiger</i>	trickling; shale	<i>seon</i> , OHG <i>sīhan</i>	ooze, trickle; filter
ON					
<i>vafra</i>	move back and forth, hover, waver	* <i>wēp/b(a)ra-</i> , OE <i>wæfre</i>	unsteady, wavering	<i>vāfa</i> , * <i>vǣf-</i>	sway, waver, wobble

Table 4: OHG and OE *r*-iteratives associated with type 2) adjectives (base *a*; *v*)

Our claim is that it was precisely in derived verbs such as the ones in Table 4 that the reanalysis of *-(V)r-* from adjectival suffix to verbal (iterative) modifier took place because the *r*-suffix was interpreted as modifying the root of the verbal base rather than as an adjectival suffix in a deadjectival verbal structure. We propose that these deadjectival verbs originally had the structure illustrated in Fig. 13, in which the root combines with a categorizing head, in this case adjectival/stativizing *a*, which is then merged with a verbalizer *v* to create a deadjectival verb stem. We further assume that inflectional class morphemes such as the marker *-ō-* of the second weak class *adjoin* to *v* rather than spelling out the head *v* itself, following the analysis of Catalan and Spanish verbal theme vowels by Oltra-Massuet (1999) and Oltra-Massuet & Arregi (2005).¹² In this context, the adjectival suffix *-ar-* was reanalyzed as a *root modifier*, modifying the root of the underived (“primary”) strong verbs that are usually found besides deverbal iteratives, such as the ones in Table 3. There is also some circumstantial morphophonological evidence for this reanalysis: Note that the root vocalism of some of the *r*-verbs in Table 3 differs from that of the base verbs and in fact reflects the root vocalism of the adjective, e.g., in OHG *ir-lungar-ōn* vs. *gi-līng-an* (adjective *lungar*), *weigar-ōn* vs. *wīg-an* (adjective *weigar*). These alternations reflect inherited, morphosyntactically conditioned root ablaut between, e.g., an old zero grade vs. *e*-grade of the root (e.g., PGmc. **un* < **ŋ* vs. *in* < **en* in *lungar* vs. *gilingan*) or between *e*-grade and *o*-grade (e.g., PGmc. **ei* > *ī* vs. **oi* > *ei* in *wīgan* vs. *weigar*). The original derivational function of these alternations became increasingly opaque in early Germanic and examples like these are few and far between, but they are precious evidence of the original derivational direction of the pattern by which iterative *-Vr-* came into being. However, there are also cases in which the verbal adjective has the same root vocalism as the primary verb it is derived from, in which case the base verb, the adjective, and the iterative all have the same root vowel (e.g., *wachēn* - *wahhar* - *wahharōn*), and after the reanalysis, the root vocalism of the iterative canonically matches that of the base verb anyway (e.g., *gangan* - *gangarōn*).

The root modifier structure we propose is given in Figure 14; note that nothing has changed in terms of the surface structure here. Figure 14 is essentially the structure of a primary (root-derived) verb, but with the twist that there is an element that modifies the interpretation of the root, namely the reanalyzed *-(V)r*-element (here, we use OHG as a stand-in for what was presumably a proto-NW-Germanic development).

¹² To simplify the exposition, we assume that this structure was in place already in OHG, but it is actually more likely that conjugational class markers such as *-ō-* were reanalyzed as adjuncts to *v* only in the course of the reanalysis of nominal and adjectival heads such as *-ar-* as verbalizers, creating the equivalent of “complex thematic suffixes” in, e.g., Ancient Greek and Latin. See Calabrese (2023) and Calabrese & Petrosino (2023) on this development in Latin and Grestenberger (2022) on the structure of complex thematic verbalizers in Ancient Greek.

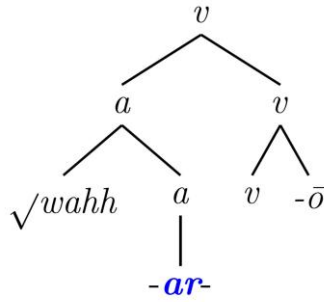


Fig. 13: Original structure of OHG deadjectival
wahharō(n)

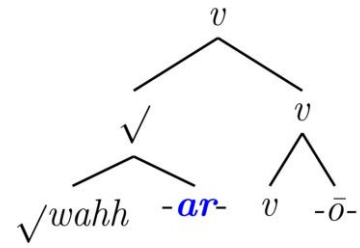


Fig. 14: OHG *-ar-* as a root modifier

The availability of such structures is motivated on the one hand by the arguments adduced in favor of treating (at least certain types of) derivational affixes as *roots* that are merged before the first categorial head is introduced into the structure, rather than as categorial heads themselves (e.g., De Belder 2011; De Belder et al. 2014; Lowenstamm 2014; Creemers et al. 2018), and on the other hand by the analysis of iterative, or rather pluractional, morphology as event modification (Kastner 2020). In Kastner’s proposal, the root that modifies the event, $\sqrt{\text{ACTION}}$, is merged in the vP . We modify this proposal by merging the modifier directly with the root of the respective verbal base to account for the linear adjacency and the fact that the compositional meaning of the resulting string is clearly different depending on what root is selected: We have already seen that $-(V)r-$ derives gradable PC adjectives when it is merged with roots expressing a state, as in ex. (7), but *verbal adjectives* when combined with eventive roots, as in ex. (8), and that the verbs built on these two different classes of adjectives likewise differ in terms of their syntax and semantics. We interpret this as supporting the view that certain roots can express events without additional syntactic structure, and that eventive roots systematically differ from stative/state roots not only in terms of their meaning but also in the way that meaning is expressed through (primary) word formation (Beavers & Koontz-Garboden 2020; Beavers et al. 2021). We actually see this difference in the older Germanic *r*-verbs as well: Recall that verbs derived from PC adjectives in $-(V)r-$ form statives and factitive-fientive alternation verbs (cf. Table 2). We have no reason to believe that the structure of these verbs differs from the one in Fig. 13, rather, the observed Aktionsart difference is the result of merging a PC root vs. an eventive root into this structure.¹³

¹³ One of the anonymous reviewers rightly fears “that there’s quite a bit of semantics leaking into roots” in this account. We agree that too rich a lexicon is undesirable and, at any rate, incompatible with DM. However, there are versions of DM in which roots do have basic denotations such as STATE, EVENT, THING, e.g., Harley 2005; Anagnostopoulou & Samioti 2014, and even more fine-grained ones splitting the event category into, e.g., ($\pm$)AGENTIVE (where AGENTIVE corresponds to our $\sqrt{\text{ACTION}}$), EXTERNALLY CAUSED, INTERNALLY CAUSED, RESULT, etc. have been proposed, e.g., Harley & Noyer 2000; Alexiadou et al. 2006; though these interpretational differences have also been treated as due to “encyclopedic knowledge” (Harley & Noyer 2000; Embick 2021: 79) rather than formal properties of the lexical entries of roots. Here, we take our findings that *some* early Germanic categorizers are sensitive to root type, together with the arguments adduced by Beavers & Koontz-Garboden (2020) and Beavers et al. (2021), as further evidence for this type of approach. However, the question of how much meaning exactly is contained in roots is a matter of ongoing and controversial debate and may be subject to cross-linguistic variation. For example, Alexiadou & Lohndal (2017) propose a gradient scale of how much content is expressed by the root vs. by the first categorizer in a given language, where Hebrew is at one end of the scale in having relatively ‘empty’ roots, while roots in English are relatively ‘contentful’, which notably coincides with different categorization requirements for roots in Hebrew vs. English. Old English falls somewhere in the middle, and the same seems to be the case for the other older NW Germanic languages discussed here (Grestenberger & Kastner 2022: 52 briefly outline a diachronic pathway for how roots could become more ‘contentful’ over time). Our analysis thus does not unnecessarily multiply the *entia* already under discussion in the literature.

As the final step of the change from adjectival *-(V)r-* to verbal “iterative” *-(V)r-*, we propose the structure in Fig. 15, in which the modifier *-(V)r-* was finally reanalyzed as verbalizing *head* (the diachronic reanalysis is again indicated by an arrow).

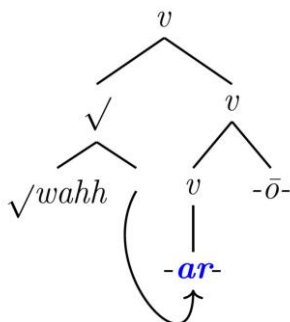


Fig. 15. OHG *-ar-* as a verbalizer

We propose that this reanalysis had taken place by the time OHG, OE, and ON are first attested. That affixes (especially prototypically modificational ones such as diminutives) can act either as heads or as adjuncts/modifiers has been argued by, e.g., Wiltschko & Steriopo (2007); Fábregas (2013); Gouskova & Bobaljik (2022), all of whom propose several phonological, morphosyntactic and morphosemantic criteria for distinguishing between affixes as heads vs. affixes as modifiers. A crucial diagnostic is that heads determine *category* and inflectional features such as inflectional class, while modifiers are not category-changing (e.g., diminutive affixes that can modify nouns, adjectives, or verbs alike) and do not determine inflectional class. In other words, the inflectional class of the base does not change. Applying these diagnostics to the older NW-Germanic languages, the fact that iterative *-ar-* in OHG is consistently associated with weak class II inflection (i.e., that it determines inflectional class in the relevant sense) and that the collapse of the verbal inflectional classes that independently took place in the older NW-Germanic languages led to the increased fusion of the verbal stem-forming suffixes with the (phonologically reduced) inflectional endings suggests to us that *-ar-* and structurally similar elements had become reanalyzed as verbalizing heads at that stage (cf. also fn. 6).

Finally, we end with a speculative note on the *semantic* development of these verbs. The structure we have argued for in Fig. 14 suggests that “iterative” is a type of event modification, a meaning that was inherited by the reanalyzed verbalizer in Fig. 15. We have not yet explained what exactly this event modification is and why it was specifically associated with verbs built on “verbal adjectives”. We speculate that this development is connected to the fact that verbal adjectives built on eventive roots — that is, type 2) adjectives such as the ones in (8) — tend to develop modal readings, at least in the languages in which these formations have been studied in depth. These readings include dispositional and deontic modality (see Hackstein 2003 for a survey).¹⁴ This is also the case for the type 2) *r*-adjectives in the older NW-Germanic languages, many of which are glossed with a dispositional meaning (‘prone to x’), e.g., OHG *wahhar* ‘(prone to be) vigilant’, *lungar* ‘capable’, *weigar* ‘prone to fight, combative’. Hackstein (2003) argues that such dispositional readings can give rise to gerundives, in which a necessity is predicated on an agent. Here, we propose to identify this modal component of verbal adjectives from eventive roots as the “agentive quale” (P[‘tendency-to’]) which Fábregas (2020) takes to be the semantic component that gives rise to dispositional readings in

¹⁴ Modern Greek verbal adjectives (that is, root-derived “adjectival participles”) in *-to-* can also have deontic modal (ability/possibility) readings, cf. Anagnostopoulou & Samioti 2013. On the properties of adjectives derived from eventive roots cf. also Anagnostopoulou 2003; Embick 2004; Anagnostopoulou & Samioti 2014; Grestenberger 2020; Beavers et al. 2021; etc.

deverbal adjectives. Although this is very tentative, we propose that this ‘tendency-to’ of adjectival *-(V)r-* was reanalyzed as something akin to Kastner’s (2020) event modifier *vaction*, predicating the tendency-to *x* (where *x* = event expressed by a root) on the subject of the modified verb. That is, under the ‘tendency-to’ modificational reading, the structure in Fig. 14 would have meant ‘tends to *wachhar*’. This reading could have given rise to the event-internal pluractionality/iterative reading of the resulting verbalizer (cf. Fig. 15) through a grammaticalization of the implicature that “tend to *x*” means “*x* happens more than once”. That the resulting verbs are unergative activity verbs would then follow from the agentive quale associated with such dispositions. However, more work is needed to expand on this proposal, including philological work on the extent to which the disposition reading of verbal adjectives is attested in the textual sources of the older NW-Germanic languages.

6. Conclusion

In this article, we have presented a corpus study of “*r*-verbs”, verbs which contain a morpheme *-(V)r-* either as part of their derivational base or as a verbal stem formant, in four older Germanic languages (Gothic, Old High German, Old English, Old Norse) in order to understand the origin of the so-called deverbal iterative class of *r*-verbs in these and their Modern Germanic descendant languages (e.g., Engl. *flicker*, *flutter*; PDG *flattern* ‘flutter’, *folgern* ‘deduce’). In line with the handbooks, we found that the deverbal/iterative *r*-verbs are a (proto-)NW Germanic innovation. Our survey of OHG, OE and ON further showed that if the *r*-segment is part of the base, the derived verb tends to be a denominal accomplishment or achievement verb if the base is nominal and a deadjectival state or degree achievement verb if the base is adjectival. If the *r*-segment is not part of the base, the derived verb tends to be a deverbal/root-derived iterative.

Since cross-categorial derivation is a crucial context for the reanalysis of categorizers (Grestenberger 2022; 2023), we have proposed that the iterative verbalizer *-(V)r-* had its origin in deadjectival verbs based on verbal adjectives from eventive (non-PC) roots formed with the adjectival suffix *-(V)r-*, which was reanalyzed first as a root modifier and then as a verbalizing head itself. We have tentatively suggested that the iterative/pluractional semantics of this suffix could have arisen from certain dispositional readings associated with adjectives from eventive roots. The reanalysis of an originally *adjectival* suffix as a *verbal* suffix thus took place in a very specific subclass of deadjectival verbs. These verbs were moreover shown to be crucially distinct from (factive/fientive) degree achievement verbs, which only saw a limited intrusion of positive or comparative *-r-* (of the type *ver-breit-er-n* ‘broaden’) and crucially no reanalysis of the positive/comparative *r*-suffix, which as suggested by cross-linguistic evidence would have given rise to a factitive or factitive-fientive verbal stem-forming suffix rather than an iterative one.

Nevertheless, many details of the semantic reanalysis remain to be further elucidated. As mentioned several times, the deverbal iteratives are distinct from denominal verbs in the older Germanic languages both morphologically in terms of inflectional class (denominals use weak classes I and II, deverbals only class II) and semantically. On the other hand, a diachronic connection between denominal activity verbs and the causative/iterative “polysemy” has also been proposed in the typological and historical literature (e.g., Kölligan 2004, 2007; Aikhenvald 2011; Peters 2016; Bozzone 2020), and it would be worth exploring to what extent (if at all) the deverbal iteratives interacted with causative/iterative readings associated with erstwhile denominals in the older Germanic languages.

In the future, we moreover plan to extend our synchronic survey to Old Saxon and Old Frisian (cf. fn. 4) on the one hand and to add Middle and Modern High German and Middle and Modern English data on the other hand, in order to further investigate the diachrony of this enigmatic verbal category across time. Specifically, the integration of denominal verbs based on nominal stems in which *-er* (OHG *-ir*) was identified as the collective plural ending of neuter nouns (Wegener 1999: 19 and the references

therein) deserves closer attention, since the resulting type became productive in MHG (e.g., PDG *ver-gött-er-n* ‘to treat as a god’ from *Gott*, pl. *Gött-er*), *blätt-er-n* ‘to leaf through’ (*Blatt* ‘leaf, page’, pl. *Blätt-er*).

The diachrony of Aktionsart-changing/-modifying morphology is generally understudied. We hope to have contributed to understanding where pluractional morphology in the verbal domain may come from diachronically, though much more work is needed to determine what other possibilities exist and how general this proposed reanalysis path is. Future studies will hopefully unearth more such analogues to “grammaticalization paths” described elsewhere in the literature for various types of morphosyntactic change in the domain of word structure changes and shed light on the development not just of iterative, but also other types of lexical aspect markers such as inchoative, factitive, or activity verbalizers.

Acknowledgements

We owe thanks to Carolin Rumppler for her contribution to the data collection. We also wish to thank Jonathan Bobaljik, Gianina Iordăchioaia, Viktoria Reiter, Gudrun Samberger, and two anonymous reviewers for *Word Structure* for their constructive comments and feedback on this article, as well as the audiences at the 21st International Morphology Meeting, the Theoretical Linguistics Colloquium at the University of Vienna, and the Linguistic Colloquium at the University of Graz where this work was presented. Finally, we gratefully acknowledge funding from the Austrian Science Fund (FWF V850-G “Verbal categories and categorizers in diachrony”, PI Laura Grestenberger & FWF P32415-G “Relational Adjectives in the History of German” (RAHiG), PI Martina Werner).

Author contribution statement

Authors 1 and 2 were responsible for funding acquisition, conceptualization and supervision of the research activities, planning and execution of the project. Author 1 co-ordinated the conceptualization and initial draft, wrote sections 1-2 and 5-6, and created the data visualization in Section 4. Author 2 was responsible for the data analysis in Section 4. Author 3 was responsible for data collection and methodology (Section 3). Author 4 contributed to the data collection and validation and to the Appendix. All authors contributed to the conceptualization of the article and commented on the final draft.

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Abbreviations

CoS = Change of State

Go. = Gothic

MHG = Middle High German

NW-Germanic = North-West Germanic

OE = Old English

OHG = Old High German

ON = Old Norse

PC = Property Concept

PDG = Present Day German

PGmc. = Proto-Germanic

PoS = Part of Speech

Appendix

Gothic

	Entry	Meaning	Base	Base-r	Base.meaning	Base.cat
1	abrjan bi-	be beside oneself about something with fear/amazement; astonish	abrs	x	severe, extreme	a
2	fahrjan ga-	prepare	fagrs	x	fit; beautiful	a
3	gaurjan	offend sb.	gaurs	x	sad, sorrowful	a
4	huggrjan	hunger, starve	hūhrus, hunhrus*, huggrus*	x	hunger, famine	n
5	kaurjan (ana-)	burden/be burdensome, complain	kaurus	x	burdensome	a
6	maurprjan	murder	maurpr	x	murder	n
7	mērjan (us-)	proclaim, praise	*mēri-; -mers	x	famous	a
8	reiran	shake	reirō	x	tremors, earthquakes	n

9	riurjan	corrupt	riurs	x	ephemeral	a
10	stiurjan	establish, affirm	*steurō-	x	regulation	n
11	tagrjan	weep, cry	tagr	x	tear	n
12	timrjan	edify, build; embolden	*timra-	x	building material, timber	n
13	ufar-hleiþrjan	to tent oneself/pitch a tent over oneself	hleipra	x	hut, tent	n
14	wajamērjan	blaspheme, slander	wajamērei	x	bad reputation	n

Old High German

	Entry	Meaning	Base	Base-r	Base.meaning	Base.cat
1	afaren, aberen	repeat, restart	afur	x	but, however	adv
2	afarōn	repeat, renew	afur	x	but, however	adv
3	āmarōn, jāmarōn	lament, complain, grieve	āmer	x	grief	n
4	antharōn, antrōn, (gi-)anterōn	imitate	*ant-haro; *ant-hara	x	imitator; imitation	n
5	argerōn, -argorōn, (gi-)	worsen, cause to deteriorate, destroy	arger, ergir	x	worse, more evil	comp
6	barōn (gi-, int-, ir-)	reveal	bar	x	bare	a
7	bez(z)erōn, pez(z)erōn	better, improve	bez(z)ero	x	better	comp
8	bismarōn	mock, ridicule	bismar, bismar	x	disdain, mockery	n
9	bismerēn	mock, ridicule	bismar, bismar	x	disdain, mockery	n
10	bismerien	mock, ridicule	bismar, bismar	x	disdain, mockery	n
11	bittaren	embitter	bittar	x	bitter, sharp, hurtful	a
12	bittarōn	be angry, bitter	bittar	x	bitter, sharp, hurtful	a
13	bliuwarōn	beat (sb. up)	bliuwan	-	beat (sb. up)	v
14	donarōn, thonarōn (bi-)	thunder	donar, thonar	x	thunder	n
15	eitaren	poison, fester	eit(t)ar, heitar, eit(t)er	x	poison, pus	n
16	ērōn	honor	ēra	x	respect, honor	n
17	fezzarōn	shackle	fezzara	x	shackles	n
18	fideren, fidiren	grow feathers, fledge	fedara	x	feather	n
19	finst(a)rēn	darken	finstar, finstir, finster	x	dark	a
20	finst(a)ren, finistren (bi-)	darken	finstar, finstir, finster	x	dark	a
21	flogarōn	flutter, flicker	fliogan	-	fly, fly around, float, glide	v
22	fluob(i)ren	comfort	fluob(a)ra	x	comfort	n
23	fordarōn, forderōn, fordorōn (ana-)	demand (accountability); accuse, prosecute	ford(a)ro	x	before, ahead, front	a
24	fuot(i)ren, fuatiren, fuataren	feed	fuoter, fūter	x	fodder	n
25	furdiren	promote, forward	furdir	x	further, forward	adv
26	galstarōn (bi-)	enchant, bewitch	galstar	x	magic, trickery, witchcraft	n

27	gangarōn	walk around, roam	gangan	-	walk	v
28	gangarōn -ana	attack	ana-gangan	-	attack	v
29	harmskarōn, harmscarōn	plague, bother, torment	harmscara	x	punishment, torment	n
30	heitaren	be bright, clear; make happy	heitar	x	bright, sunny, cheerful	a
31	hēren (gi-, int-), (sih)	rise up, make oneself great, honour (oneself), int-: disdain	hēr	x	great, venerable, noble	a
32	hungiren, hung(e)ren	hunger	hunger, hungir, hungar	x	hunger	n
33	last(a)rōn	gossip, blaspheme	lastar	x	insult	n
34	līhterōn, liehterōn (ir-)	lighten; mitigate	līht(i), comp. līhter	x	light	comp
35	loubirōn	become lush, green, leafy	loubi, pl. -ir	x	leaves, foliage	n (pl)
36	lōwirōn	tan	lō	-	bark, tanbark	n
37	lungarōn ir-	roam, wander around, loiter	lungar; gi-lingan	x/-	spry, capable; succeed	a; v
38	lūttaren (gi-, ir-)	purify, clarify, clean, explain	lūt(t)ar, hlūt(t)ar	x	pure, clear	a
39	lūttarēn, hlūttarēn	glow, shine, be bright	lūt(t)ar, hlūt(t)ar	x	clear, bright	a
40	magaren	lose weight, become meager	magar	x	meager, skinny	a
41	mar(r)ēn, mären	make famous, announce	māri	x	famous	a
42	mērōn	increase	mēro, mēra; mēr	x	more	comp
43	minnerōn	reduce, diminish, decrease	minniro; adv. minnira	x	smaller, lesser	a; comp., adv
44	munt(a)ren, (ir-)	awaken, wake up	muntar	x	awake	a
45	murdiren	murder, slay	murd(i)ro; murden, morden;	x/-	murderer; to murder	n; v
46	nidiren, nider(r)en	degrade, humiliate	nidar, nider, nidir	x	below, under	a; prep., adv
47	obarōn (gi-)	delay, inhibit, stall, overcome (*'be higher, upper')	obaro	x	upper, higher	a
48	purperōn, purpurōn	dress in purple	purpura	x	purple	n
49	rīt(e)rōn	sift, strain	rītara	x	sieve	n
50	ruodarōn	row, move by using oars	ruodar	x	oar	n
51	scaberōn, scaborōn, skabarōn ir-	scratch together; catch, win	skabar; (gi-)scaban, -en	x/-	rough, scabby, scratchy; itch, scratch	a; v
52	sērēn	feel pain	sēr	x	painful	a
53	sigirōn (ubar-)	defeat	sigis (*s-st.)	(x)/-	victory	n
54	skarōn (gi-)	form a host, bring into formation	skara	x	flock, crowd, host	n
55	smeckarēn	consume oneself, pine away	smehhar; (gi)smecken; (gi)smackēn	x/-	delicate, refined; taste, savor; taste of	a; v
56	smekarōn, -smehharōn (gi-, un-)	refine, make delicate (un-: unrefined)	smehhar; (gi)smecken; (gi)smackēn	x/-	delicate, refined; taste, savor; taste of	a; v
57	snuoren	string, lace up	snuor	x	string	n
58	sparēn	save, maintain, preserve	spar	x	sparse, scarce	a

59	sparōn	save, maintain, preserve	spar	x	sparse, scarce	a
60	spirdaren (ana-, ingagan-, widar-, zuo-)	toil, struggle, stretch, support	?	?	?	?
61	stabarōn	vibrate, sway, twitch	(ir-)stabēn	-	be(come) stiff, rigid	v
62	starēn	stare	stara°	x	stiff, rigid, fixed	a
63	starōn	stare	stara°	x	stiff, rigid, fixed	a
64	stegaren, stegeran	ascend, rise	stegōn	-	climb, ascend	v
65	stiuren, stiurran, stuorren	steer, lead, guide	stiura	x	helm, rudder	n
66	stobarōn	be struck with, wonder	*steup- (cp. Lat. (ob)stupere)	?	?	?
67	sūbarōn	clean	sūbar, sūb(i)ri	x	pure, clean	a
68	sūbiren, sūb(e)ren	clean	sūbar, sūb(i)ri	x	pure, clean	a
69	suntarōn	separate, eliminate	suntar; suntaro	x	separate	adv; a
70	swangaren	impregnate	swangar	x	pregnant	a
71	swārōn, swarōn (be-)	press heavily on, weigh down; complain	swār	x	heavily	a
72	swebarōn	float, swim, drift on the water	swebōn	-	float, wave, swim	v
73	tantarōn	speak drivel, babble, ramble	unclear; maybe related to MHG <i>tant</i> 'frippery; drivel'	?	MHG <i>tant</i> 'frippery; drivel'?	? (-/n)
74	timberēn (bi-)	turn murky, dull	timber	x	dark	a
75	timberen, timberen (bi-)	become dark, blur	timber	x	dark	a
76	uoberōn	examine, exercise, practise	uoben	-	exercise, practise	v
77	wacharōn	wake, watch over	wachar; wāhhēn	x/-	awake, aware	a; v
78	warōn (bi-)	pay attention, consider	wara; (gi)war	x	observation, perception, attentive	n; a
79	weigerōn, weigarōn	reject, refuse	weigar; weigen	x/-	refuse, tire	a; v
80	widerōn, widarōn, widorōn	reject, resist, compete	widar	x	against	prep, adv
81	wimmeren	swarm, teem	wium(m)en, wimmen	-	swarm, scurry	v
82	wirserōn	worsen, make worse; press down	wirsero, wirsero; wirsōn	x/-	worse; press down, oppress	comp; v
83	wirsiren	worsen, make worse; press down	wirsero, wirsero; wirsōn	x/-	worse; press down, oppress	comp; v
84	wuntaren	wonder, be amazed	wuntar	x	wonder	n
85	wuntarōn, wont(a)rōn	marvel at, wonder	wuntar	x	wonder	n
86	wuohherēn	grow	wuohhar, wuohhir, wuohher, wuahhar	x	growth, harvest, yield	n
87	wuohherōn	grow, proliferate, grow rampant	wuohhar, wuohhir, wuohher, wuahhar	x	growth, harvest, yield	n
88	zimb(e)ren	build, construct	zimber, zimbar	x	material, fabric, construct	n
89	zimberōn, zimb(o)rōn,	build, construct	zimber, zimbar	x	material, fabric, construct	n

90	zoubarōn, zouverōn (bi-)	bewitch	zoubar	x	magic, witchcraft	n
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Old English

	Entry	Meaning	Base	Base-r	Base.meaning	Base.cat
1	ættrian, ætrian (ge-)	poison, envenom	attor, ātor	x	poison	n
2	beterian	make better, improve	betre	x	better	comp.a
3	biterian (a-)	make bitter, embitter	biter	x	bitter	a
4	bysmorian, bysmerian	insult, mock, deride	bismer, bysmer	x	mockery, insult	n
5	ceorian, ceorian (be-)	complain	caru	x	concern, worry	n
6	clifrian	scratch	clifer ;clifian	x/-	claw, talon; cleave, adhere	n; v
7	derian	hurt, harm	daru	x	harm, damage	n
8	dyderian (be-)	deceive, delude	*dūðian (fries. doddjan 'tumble')	-	move, tumble	v
9	dyran	hold dear	dýre	x	excellent, noble, honoured	a
10	fægrian (a-)	become/make (something) beautiful, beautify, adorn	fæger; fægerra	x	beauty, fairness; beautiful	n; a
11	feterian (ge-)	bind	fetor, feter	x	fetter	n
12	feþrian, fiþrian	become feathered, grow feathers, flutter	feþer, feðer	x	feather, wing	n
13	fiþerian, fiðerian, fiðerigan, fiðrian (ge-)	provide with wings	feþer, feðer	x	feather, wing	n
14	flicerian, flicorian	flutter, hover,	flēogan	-	fly	v
15	floterian	flutter, float to and fro	flotian	-	swim, trickle	v
16	frēfran	console, comfort	frōfor	x	comfort, consolation	n
17	frēfrian	console, comfort	frōfor (frefer?)	x	comfort, consolation	n
18	fyrþrian (ge-)	further, advance, support	furþra; furðor, forðor	x/x	further	comp; adv
19	gadrian (ge-)	gather; be in agreement	geador	x	together	adv
20	geōmrian, gíemran	murmur, grumble, complain	geōmor	x	sad	a
21	glæterian	shine, gleam	glæd(-) (OHG glat)	-/-	bright, shiny; shine, gleam	a; v
22	hārian	become grey(-haired) or hoary	hār	x	grey, hoary	a
23	hindrian	hinder, impede, obstruct	hinder	x	behind, back	adv
24	hlēoprian	make a noise, sound	hleopor	x	sound, noise	n
25	hluttrian, hlūtrian	purify, cleanse/become clarified	hluttur, hlūtor	x	pure, clean	a
26	horian, horwian, horgian ge-	spit upon, defile	horh, horu	x	phlegm	n
27	hrīdrian, hriddrian	sift	hrīder	x	sieve	n

28	leahtrian	charge with crime, impeach, accuse, blame,	leahter	x	crime	n
29	lēþrian, lȳþrian	anoint, wash, lather	leaþor	x	lye soap	n
30	mægerian	make meager	mæger	x	meager	a
31	mæran	make famous, praise	mære	x	famous, famed	a
32	mæþrian	show respect to, honour	mæþ; mæþian	-/-	measure; regard, respect	n; v
33	martyrian	martyr	martyr	x	martyr	n
34	myrþrian (a-, for-, of-)	murder	myrþra	x	murder	n
35	niperian, nīþrian	bring low, humiliate	nīþer	x	down, below	adv
36	oferian, uferian (ge-)	raise, exalt	ofer	x	upward, above	prep; adv
37	scimrian, scȳmrian, scīmrian, scīmerian, etc.	glitter, glitter, shine, shimmer	scīma; scīmian	-/-	brightness, light; shine, gleam	n; v
38	scīran (a-, ge-, be-)	get/make clear, clarify	scīr	x	clear	a
39	searian (for-, a)	grow withered, pine away/make (something) dry, withered	seár	x	dry, withered	a
40	seolfrian (be-, ofer-)	cover with silver	sioluf, seolf, siolfor	x	silver	n
41	sicerian	trickle, sink down	*sicor (cf. MHG seiger); seon	x/-	trickling, shale; ooze, trickle	a; v
42	sigorian, sīgerian, sigrian	vanquish	sīgor	x	victory	n
43	slid(o)rian	slip, slither	slidor; slidan	x/-	smooth, slippery; slide, slip	a; v
44	sparian (a-, ge-)	spare, preserve	spær	x	sparse	a
45	spyrian (a-, aefter-, be-, ge-, of-)	search, explore	spor	x	trace	n
46	stamrian, stamerian	stammer	stamor	x	stammering	a
47	stancrian	sprinkle	stencan	-	scatter	v
48	starian (ge-)	stare	stær(e)°	x	stiff, vacant (look)	a
49	stīeran (ge-)	steer	stēor	x	guidance	n
50	sūrian (a-)	be/turn sour	sūr	x	sour	a
51	sweðrian, swiðrian, sweoðerian; sweþrian, swæþorian, swaþorian, swaþrian, swiþrian	retire, withdraw, abate, subside	unclear	?		?
52	syndrian, sundrian, syndran (ge-, to-, a-, on-)	divide, separate, keep apart	sundor	x	apart	adv
53	taefran, tiefran, tīfran (a-)	(mark with) tiver	teafor	x	paint, tiver	n
54	tealtrian	shake, totter, stagger	tealtian	-	be unsteady, shake	v
55	tīdrian (a-, ge-)	grow weak, infirm	tīeder, tīdre	x	weak, tender	a
56	týdran, tȳdrian (a-, on-)	propagate, bring forth; itr. be prolific	tūdor, tudder	x	offspring, fruit	n
57	pēostrian, pȳstrian (a-, for-, fore-, ge-, of-)	make/become dark, darken, obscure	pēóstre, péostor	x	dark	a

58	þocrian	run to and fro	*þocian (Icel. þoka 'move')	-	move	v
59	þotrian	howl, wail, cry out	þeótan	-	howl like a wolf	v
60	þunrian (ge-)	thunder	þunor	x	thunder	n
61	þwárian, þwærian (ge-, mid-)	bring into agreement, make harmonious	þwære	x	gentle, pleasant, agreeable	a
62	þweorian, þwyrian	be opposed to, adverse to	þweorh	x	adverse, opposed, cross, angry	a
63	wæterian (ge-)	water, supply with water	wæter	x	water	n
64	warian (be-, ge)	to be/make aware, warn, caution, guard	wær	x	aware, alert	a
65	wederian (ge-), wiðerian	be (good/bad) weather	weder	x	weather	n
66	witran	make certain, inform	witer	x	wise	a
67	wiþerian	be against, opposed to, resist, struggle against; make hostile	wiþer(e)	x	against	adv
68	wuldrian (a-; ge-)	glorify	wuldor	x	glory	n
69	wundrian (a-, ge-, of-)	wonder at, admire; make wonderful, magnify	wundor	x	wonder, dread, monstrosity	n

Old Norse

	Entry	Meaning	Base	Base-r	Base.meaning	Base.cat
1	angra	anger, grieve, regret	angr	x	grief, sorrow, anger	n
2	blaðra	move one's tongue, ramble on, stutter	blað	-	leaf, blade	n
3	blakra	flutter, blink (one's eye)	blaka	-	flutter, wave, flap	v
4	blístra	whistle	*blēs(t)-, cf. blāsa	-	blow, breathe	v/root
5	bogra	walk stooped, bow, grovel	buga	-	shrink, give way	v
6	eiskra	roar, rage	eisa, *eiska	-	rush forth, flare, foam	v
7	eitra	poison	eitr	x	pus; frenzy	n
8	fatra	obstruct, prevent, hinder	fatr	x	obstruction, constraint	n
9	fegra	beautify, make fair	fagr	x	beautiful, fair	a
10	fjotra	fetter sb.	fjoturr	x	fetter	n
11	flaðra	fawn on, flatter, stammer	(inherited)	-	?	v/root(?)
12	fleipra	babble	fleipa; flípr, fleipr	-/x	babble; babble, lip	v; n
13	flōgra	flutter, tumble	*flag-; cf. also flakka	-	flicker, flutter, fluctuate	v/root
14	flōkra	roam around, flutter	flakka	-	flutter, fluctuate	v
15	fundra	ponder	*funda (OSwed. funda)	-	muse, ponder, find out	v
16	gambra	brag	*gambra- (OHG gambar)	x	powerful, strong	a
17	gildra	trap	gildra	x	trap	n
18	gjōgra	wander, sway, wobble	gag- (cf. gaghals, Icel. gjögur 'gap, cave, jut')	-	bend, jut (?)	?

19	glamra	rattle, be noisy	glama	-	talk animatedly, berate	v
20	gleipra	babble, chatter	gleipa	-	babble, chatter	v
21	glitra	shine, gleam, glisten	glita; glit	-/-	glitter, shine	v; n
22	glumra	make noise, yell	glumr, glaumr	-	noise	n
23	glutra	squander	glutr	x	waste, sqander	n
24	gnōllra	growl	gnella	-	bark, scream	v
25	gnōtra	rattle, clatter, crunch	gnata; gnat	-/-	clash noisily; clash, noise	v; n
26	hindra	hinder; refl. halt, linger	*hindar (comp. hindri)	x	behind, after, later	adv
27	hjūfra	lament	*hjūfan (go. hiufan)	-	wail, whine, lament	v
28	hokra	creep, crawl	hoka, hūka	-	crouch, squat	v
39	hungra	hunger	hungr	x	hunger	n
30	hýra-st	be/become happy, friendly	hýrr	x	happy, cheerful	a
31	iðra	annoy, make regret; refl. repent, regret	iðr, iðrar (inr-) or to Go. <i>idreigo</i> 'repentance'	x	entrails, bowels; repentance	n
32	jōtra	ruminate	*iðruh(t)-	-	ruminate	v/root
33	klembra	clamp, press	klōmbr	x	clip, vice	n
34	klifra	climb	klīfa	-	climb	v
35	leyðra	wash	lauðr	x	soap, foam	n
36	lōggra	wag	*lug/luk; loka	-	be limp, hang limply	v/root
37	lūðra	creep, wind, sneak through	*luðra-, *(s)lūþ- (cf. lydda 'slacker')	x/-	(be) slack, limp, useless, slacken	a; v
38	mæra	praise	mærr	x	famous	a
49	megra	make meagre, emaciate	magr	x	meagre, skinny	a
40	mūra	build, wall	mūrr	x	wall	n
41	niðra	lower, humiliate	niðr	x	down, downwards	adv
42	ofra	raise, lift up	ofar	x	(further) up, upwards	adv
43	rýra	reduce, lessen	rýrr	x	insufficient, reduced, less	a
44	særa	wound	sārr; sār n.	x/x	sore, aching; wound	a; n
45	sautra	slurp, suck through the teeth	*saut-? unclear	?	?	?
46	sigra	win, be victorious	sigr; *segiz-	x(/-)	victory	n; n
47	sindra	spark, sparkle	sindr	x	cinder, spark	n
48	skīra	purify, baptize	skirr	x	clear, pure	a
49	skjalfra	shake, tremble	skjalfa	-	shake, tremble	v
50	skotra	push, shove	skota	-	push, shove	v
51	skvaldra	babble, chatter	skvala-	-	yell, shout	v
52	slātra	slaughter	slātr	x	meat (i.e., slaughtered stuff)	n
53	slauðra	drag oneself along, drag oneself forward	slōð; *slaud-?	-	track, path; slacken, drag	n; v/root
54	slīðra	put sth. in the scabbard	slīðr	x	scabbard, sheath	n (pl?)
55	sloðra	drag oneself forward	slōð; *slaud-?	-	track, path; slacken, drag	n; v/root
56	slōðra	drag	slōð; *slaud-?	-	track, path; slacken, drag	n; v/root
57	snoðra, snuðra	sniff, snuffle	snyðja	-	snuffle	v
58	snœra	lace, string, coil	snœri	x	rope, string	n
59	snotra	make wise	snotr	x	wise	a
60	snytra	(-sk) grow wise	snotr	x	wise	a

61	spara	save, spare	sparr	x	sparing, sparse, frugal	a
62	stakra	stagger, stumble	staka	-	stagger, push, (make) stumble	v
63	stara	stare, gape	starr	x	stiff, rigid	a
64	stīra	stare	*stī-?; *stīra- (adj., cf. EFr. stīr)?	-/x	?; stiff (?)	root; a
65	stoera	augment, increase, swell	stōrr	x	big, great, strong	a
66	stumra	stumble	stamr; *stum-/stam-; *stum-ra-	-/-(x)	stammering; stammer, stumble	a; v (a)
67	stýra	steer, control, govern	*steura-, stýri	x	steering rudder	n
68	sundra	break asunder, cut into pieces	sundr	x	asunder	adv
69	sviðra	burn, singe	sviða, sviðna	-	burn, hurt; be burned	v
70	svimra	be dizzy	svīmi; *swim(m)- (svima, swimma)	-/-	dizziness; sway, wobble, swim	n; v/root
71	taufra	enchant, bewitch	taufr (n.pl. & f.)	x	magic, sorcery, witchcraft	n
72	tendra	light, inflame	tenda, *tindan; tandr	-/x	light, inflame; fire	v; n
73	timbra	timber, build from wood	timbr	x	timber, wood	n
74	tjōðra	fetter, find	tjōðr	x	shackle, fetter	n
75	vafra	move back and forth, sway	vāfa, *vāf-	-	sway, waver	v
76	vara	warn, (cause to) be attentive	varr	x	alert, attentive	a
77	vāra	become spring	vār	x	spring	n
78	vekra	bring to life, animate	vakr	x	awake, aware, alert	a
79	viðra	blow, storm	veðr	x	weather	n
80	vitra	announce, notify	vitri; vita	x/-	wise, knowing; show, (make) perceive, recognize	a; v

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